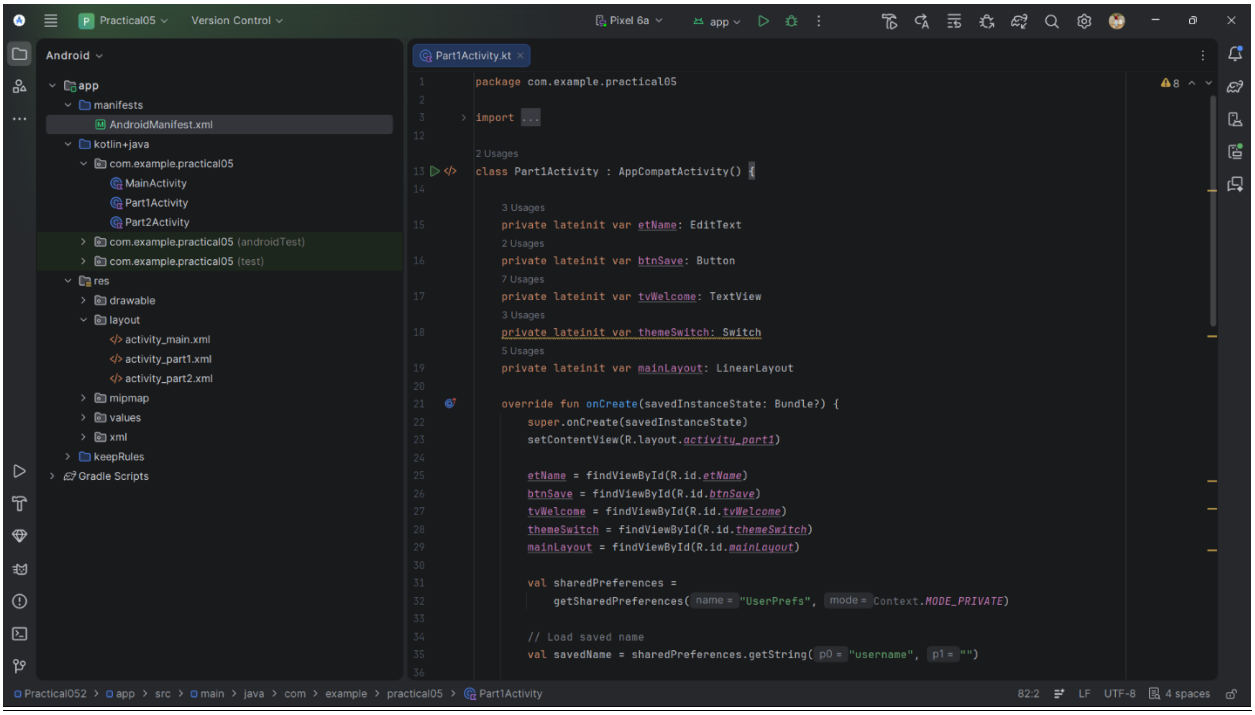
Nisandi Disara - 23001951

SCS3311 - Mobile Application Development

Practical – 05

Part 1: Shared Preferences



```
package com.example.practical05

import ...

2 Usages
class Part1Activity : AppCompatActivity() {

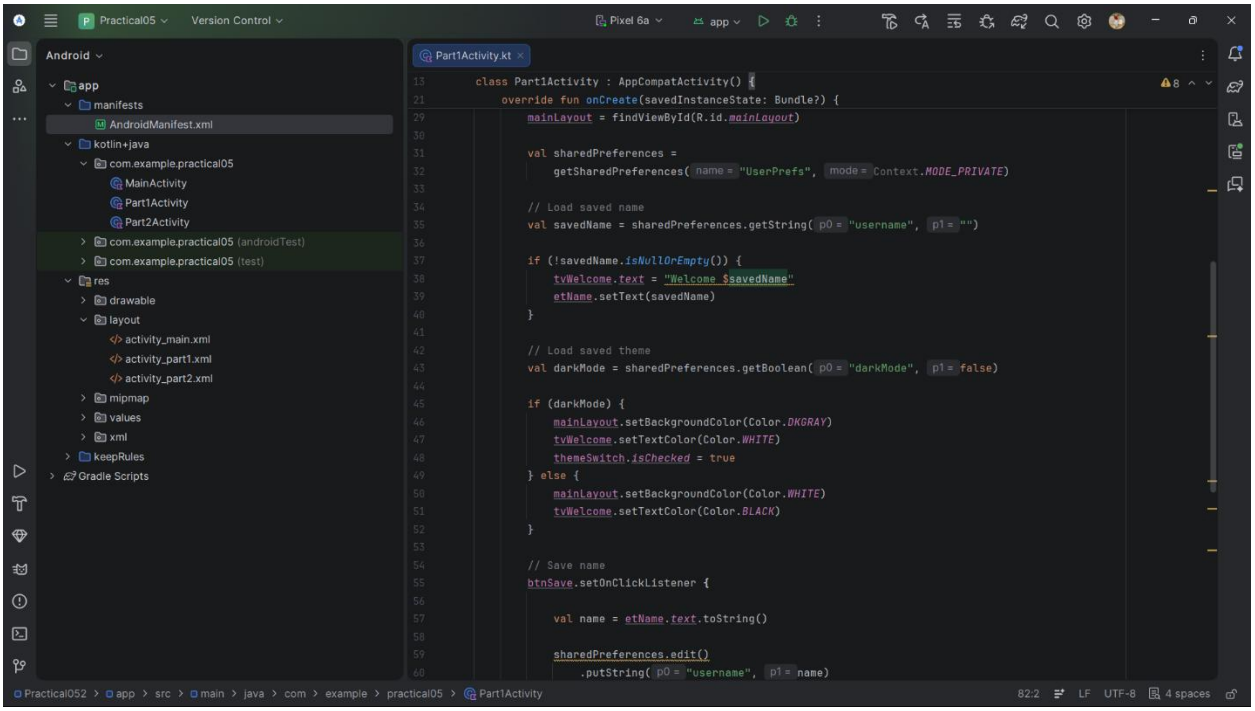
    3 Usages
    private lateinit var etName: EditText
    2 Usages
    private lateinit var btnSave: Button
    7 Usages
    private lateinit var tvWelcome: TextView
    3 Usages
    private lateinit var themeSwitch: Switch
    5 Usages
    private lateinit var mainLayout: LinearLayout

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_part1)

        etName = findViewById(R.id.etName)
        btnSave = findViewById(R.id.btnSave)
        tvWelcome = findViewById(R.id.tvWelcome)
        themeSwitch = findViewById(R.id.themeSwitch)
        mainLayout = findViewById(R.id.mainLayout)

        val sharedPreferences =
            getSharedPreferences( name = "UserPrefs",  mode = Context.MODE_PRIVATE)

        // Load saved name
        val savedName = sharedPreferences.getString( p0 = "username",  p1 = "")
```



```
class Part1Activity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        mainLayout = findViewById(R.id.mainLayout)

        val sharedPreferences =
            getSharedPreferences( name = "UserPrefs",  mode = Context.MODE_PRIVATE)

        // Load saved name
        val savedName = sharedPreferences.getString( p0 = "username",  p1 = "")

        if (!savedName.isNullOrEmpty()) {
            tvWelcome.text = "Welcome $savedName"
            etName.setText(savedName)
        }

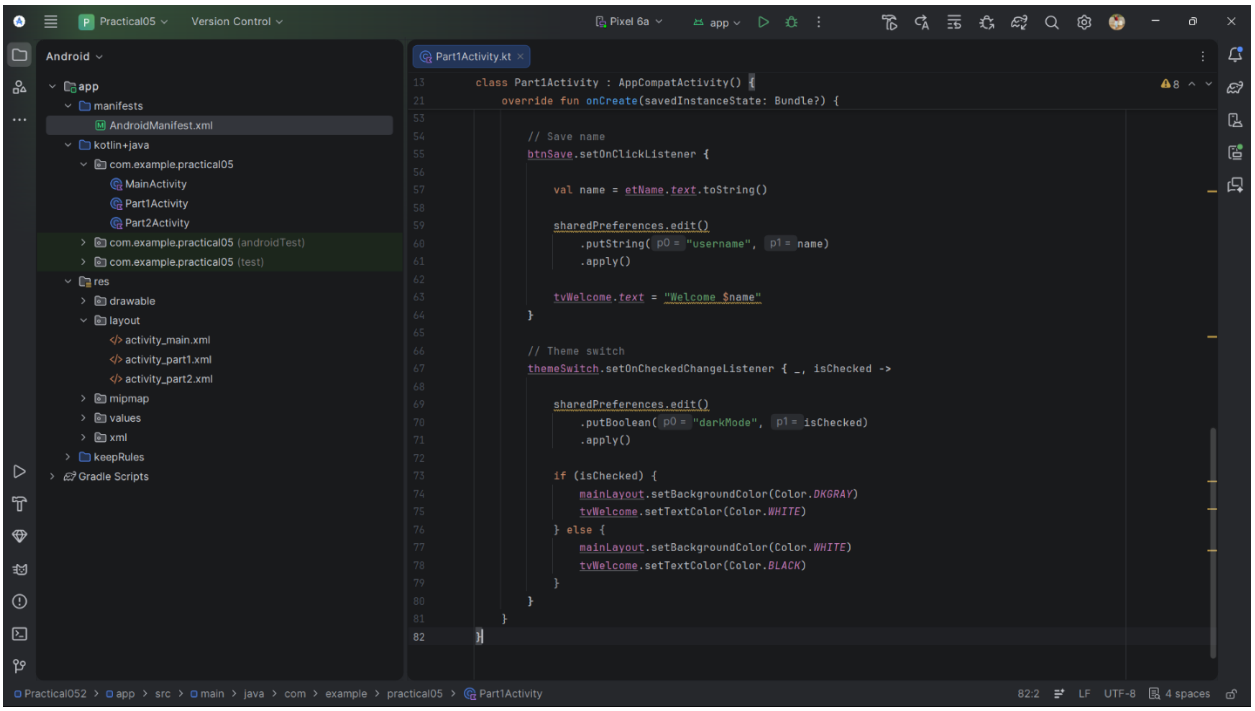
        // Load saved theme
        val darkMode = sharedPreferences.getBoolean( p0 = "darkMode",  p1 = false)

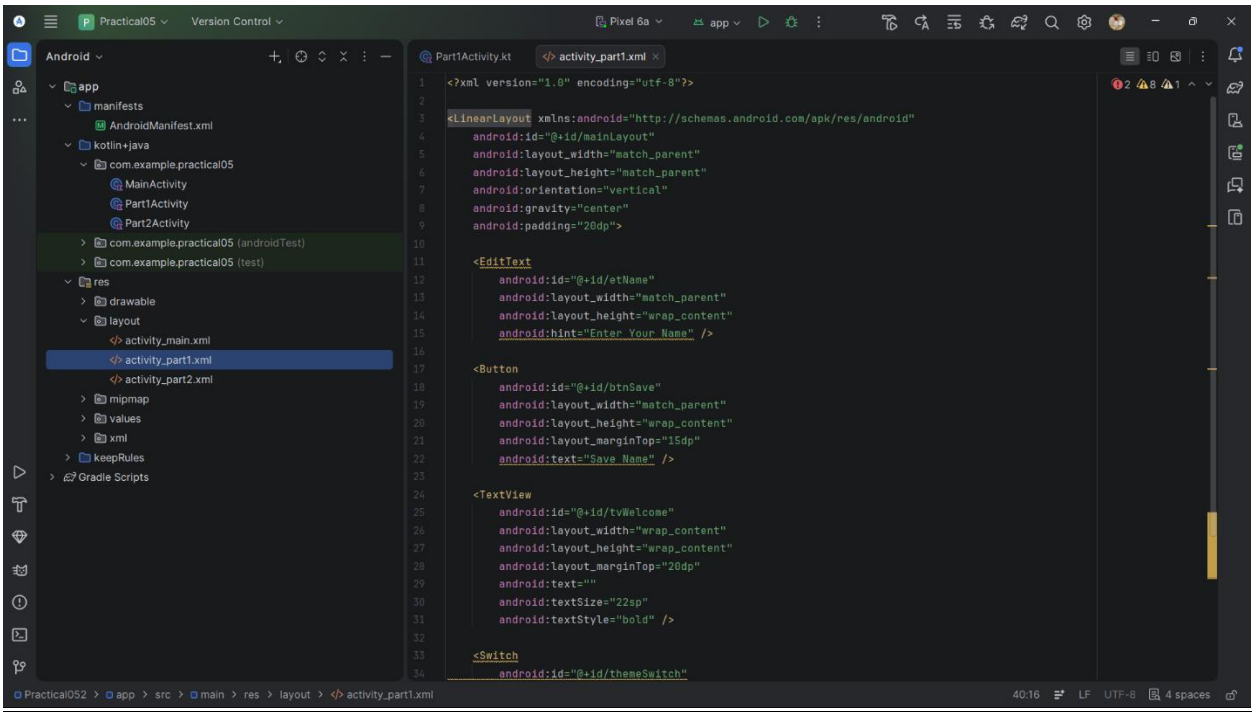
        if (darkMode) {
            mainLayout.setBackgroundColor(Color.DKGRAY)
            tvWelcome.setTextColor(Color.WHITE)
            themeSwitch.isChecked = true
        } else {
            mainLayout.setBackgroundColor(Color.WHITE)
            tvWelcome.setTextColor(Color.BLACK)
        }

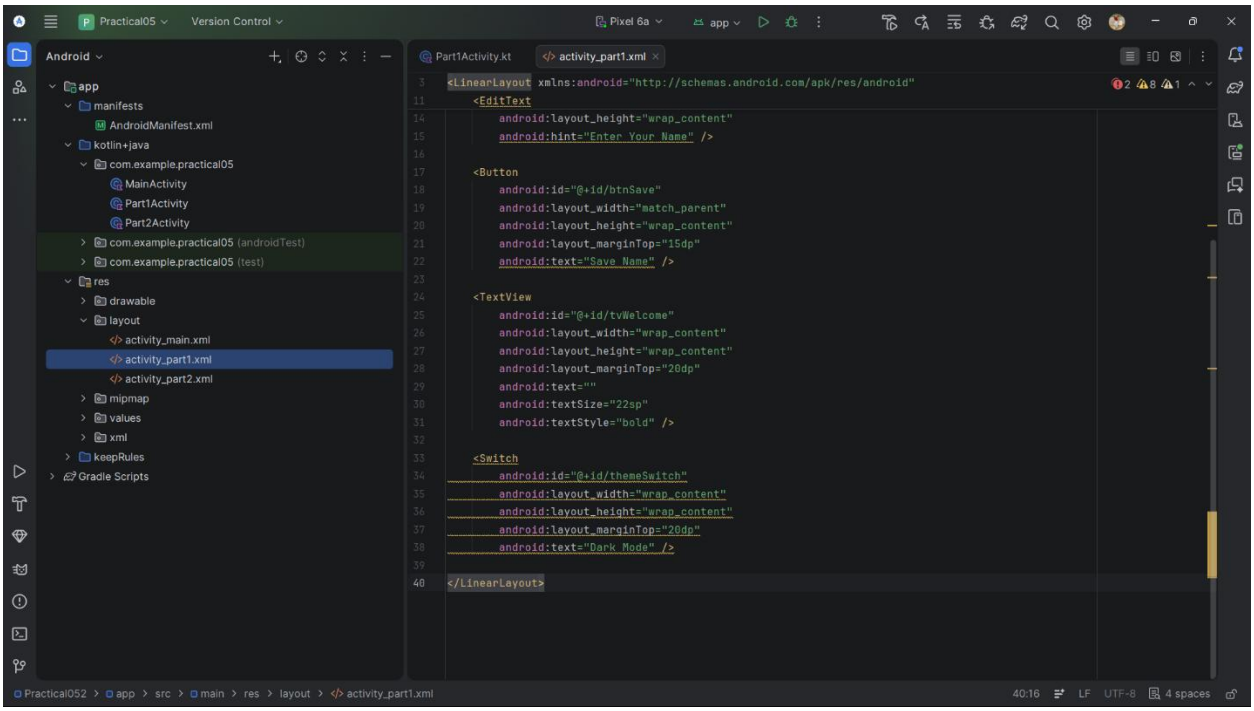
        // Save name
        btnSave.setOnClickListener {

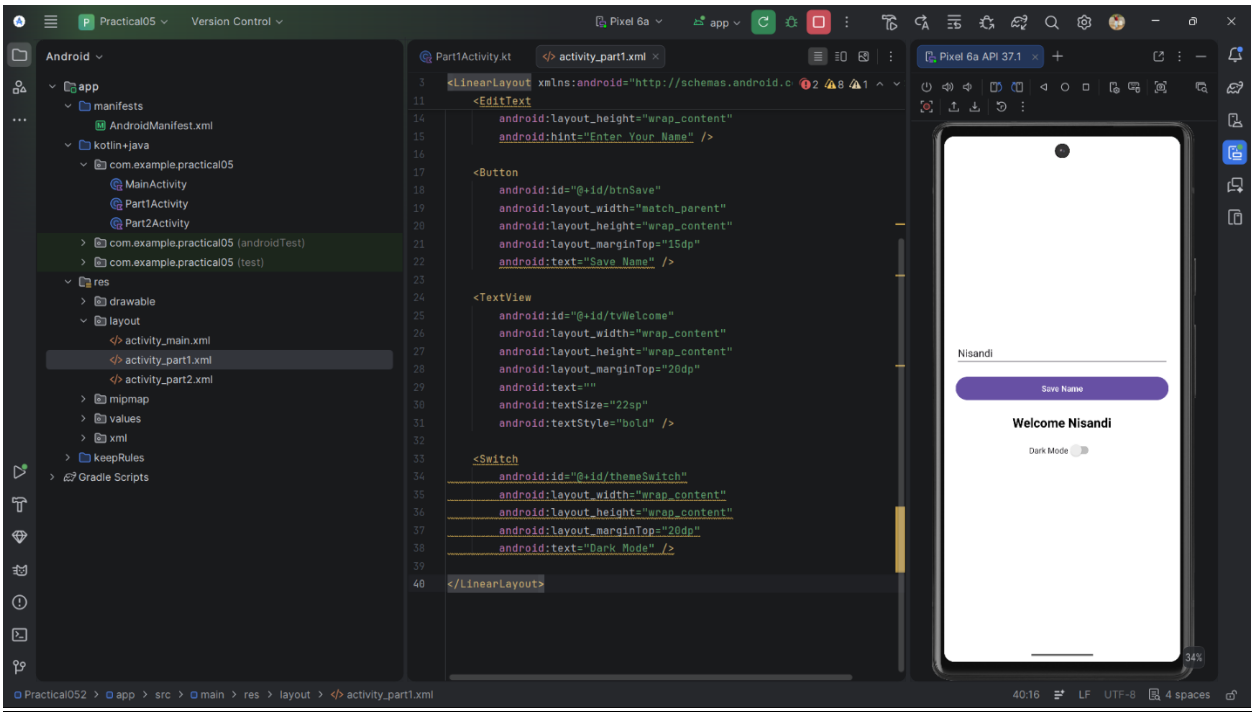
            val name = etName.text.toString()

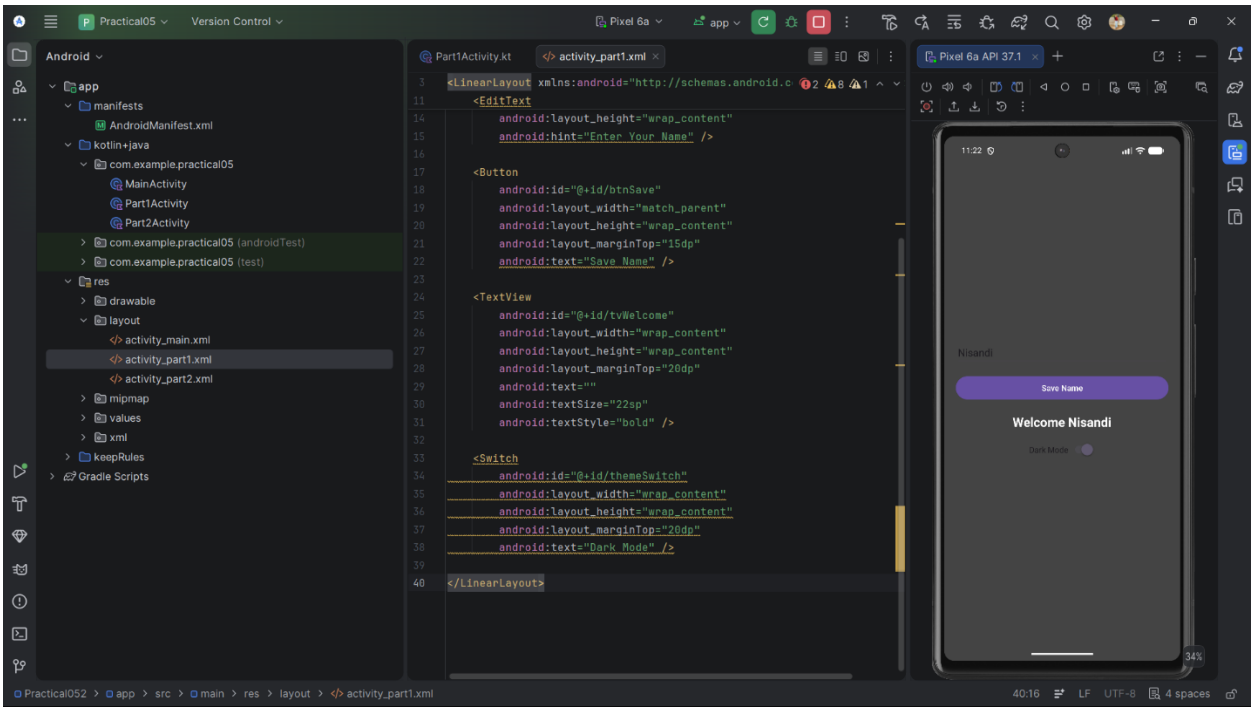
            sharedPreferences.edit()
                .putString( p0 = "username",  p1 = name)
```


Practical05
Version Control
Pixel 6a
app
Android
Part1Activity.kt
app
manifests
AndroidManifest.xml
kotlin+java
com.example.practical05
MainActivity
Part1Activity
Part2Activity
com.example.practical05 (androidTest)
com.example.practical05 (test)
res
drawable
layout
activity_main.xml
activity_part1.xml
activity_part2.xml
mipmap
values
xml
keepRules
Gradle Scripts
class Part1Activity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
// Save name
btnSave.setOnClickListener {
val name = etName.text.toString()
sharedPreferences.edit()
.putString(p0 = "username", p1 = name)
.apply()
tvWelcome.text = "Welcome $name"
}
// Theme switch
themeSwitch.setOnCheckedChangeListener { _, isChecked ->
sharedPreferences.edit()
.putBoolean(p0 = "darkMode", p1 = isChecked)
.apply()
if (isChecked) {
mainLayout.setBackgroundColor(Color.DKGRAY)
tvWelcome.setTextColor(Color.WHITE)
} else {
mainLayout.setBackgroundColor(Color.WHITE)
tvWelcome.setTextColor(Color.BLACK)
}
}
}
}
Practical052 › app › src › main › java › com › example › practical05 › Part1Activity
82:2
LF
UTF-8
4 spaces

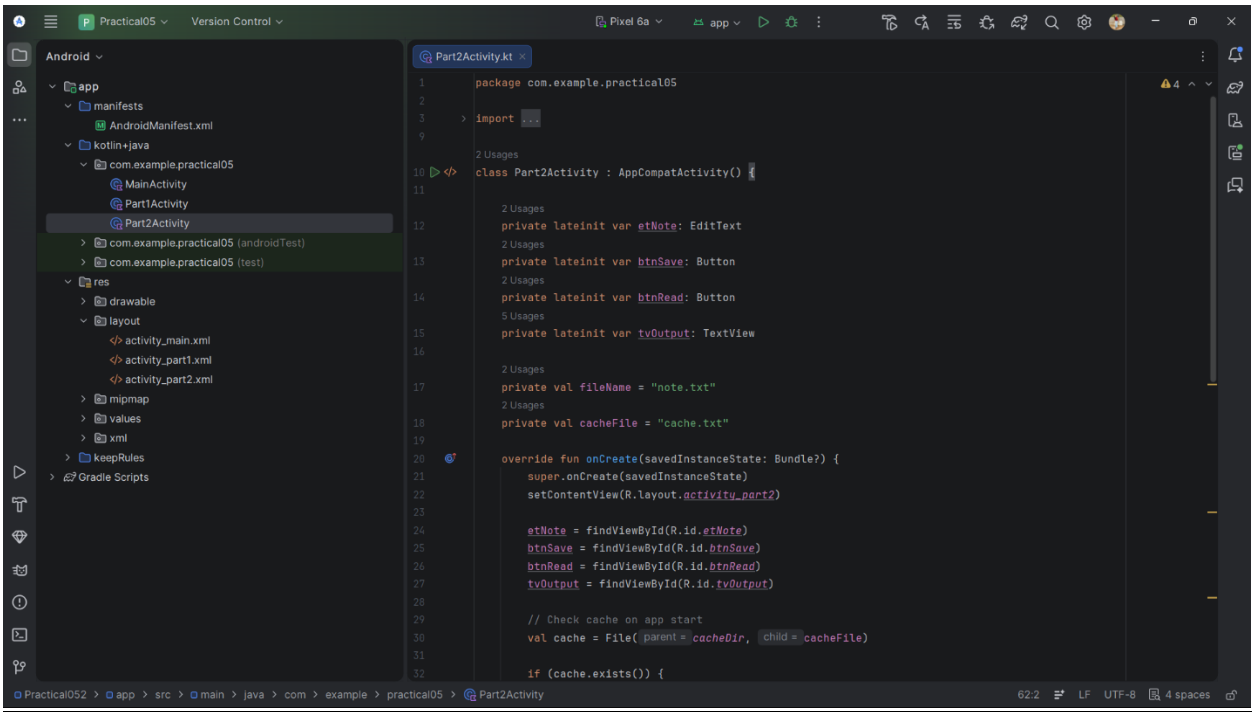

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Android
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activity_part1.xml
app
manifests
AndroidManifest.xml
kotlin+java
com.example.practical05
MainActivity
Part1Activity
Part2Activity
com.example.practical05 (androidTest)
com.example.practical05 (test)
res
drawable
layout
activity_main.xml
activity_part1.xml
activity_part2.xml
mipmap
values
xml
keepRules
Gradle Scripts
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:id="@+id/mainLayout"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:gravity="center"
android:padding="20dp">
<EditText
android:id="@+id/etName"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter Your Name" />
<Button
android:id="@+id/btnSave"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_marginTop="15dp"
android:text="Save Name" />
<TextView
android:id="@+id/tvWelcome"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="20dp"
android:text=""
android:textSize="22sp"
android:textStyle="bold" />
<Switch
android:id="@+id/themeSwitch"
Practical052 › app › src › main › res › layout › activity_part1.xml
40:16
LF
UTF-8
4 spaces

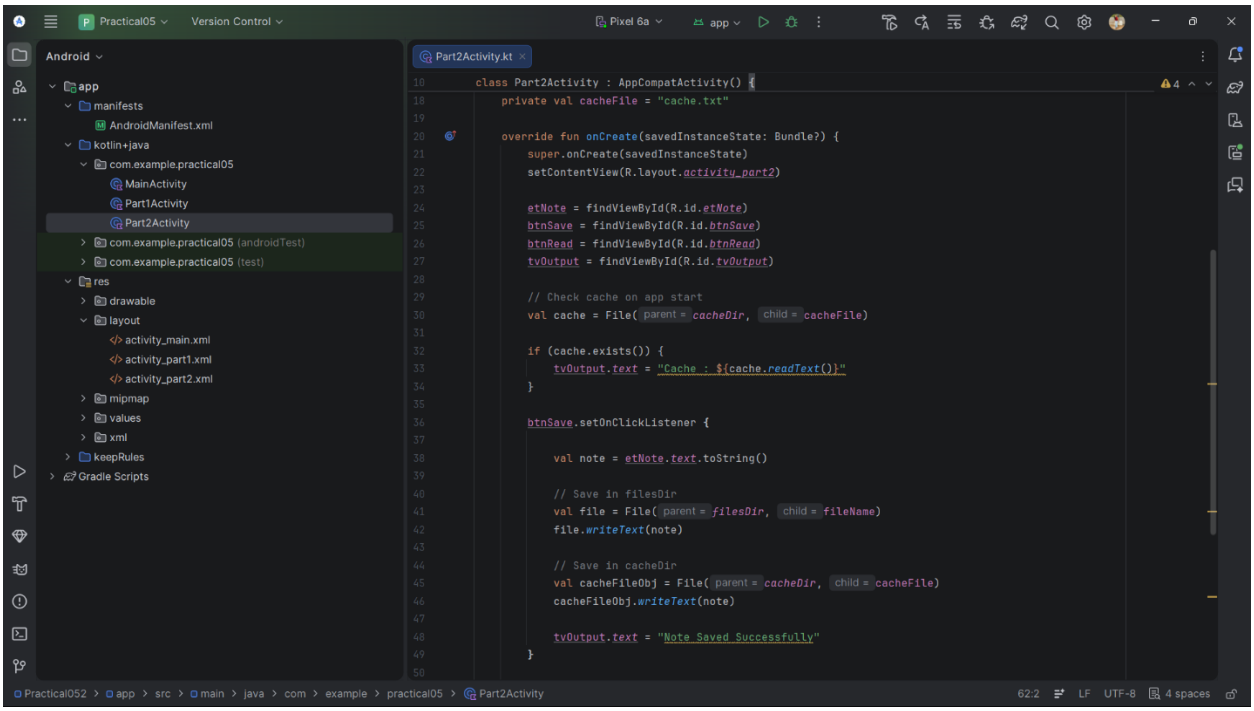

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Pixel 6a
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Android
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activity_part1.xml
app
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com.example.practical05
MainActivity
Part1Activity
Part2Activity
com.example.practical05 (androidTest)
com.example.practical05 (test)
res
drawable
layout
activity_main.xml
activity_part1.xml
activity_part2.xml
mipmap
values
xml
keepRules
Gradle Scripts
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
<EditText
android:layout_height="wrap_content"
android:hint="Enter Your Name" />
<Button
android:id="@+id/btnSave"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_marginTop="15dp"
android:text="Save Name" />
<TextView
android:id="@+id/tvWelcome"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="20dp"
android:text=""
android:textSize="22sp"
android:textStyle="bold" />
<Switch
android:id="@+id/themeSwitch"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="20dp"
android:text="Dark Mode" />
</LinearLayout>
Practical052 > app > src > main > res > layout > activity_part1.xml
40:16
LF
UTF-8
4 spaces

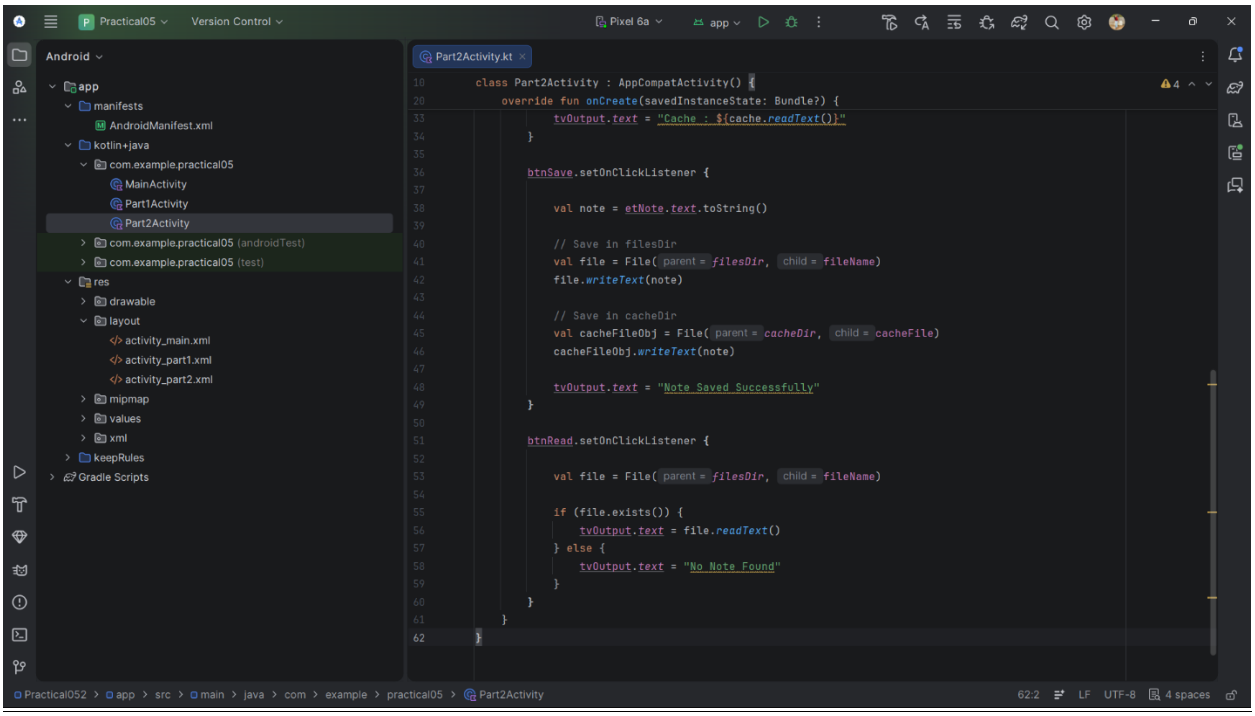

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Android
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activity_part1.xml
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manifests
AndroidManifest.xml
kotlin+java
com.example.practical05
MainActivity
Part1Activity
Part2Activity
com.example.practical05 (androidTest)
com.example.practical05 (test)
res
drawable
layout
activity_main.xml
activity_part1.xml
activity_part2.xml
mipmap
values
xml
keepRules
Gradle Scripts
<LinearLayout xmlns:android="http://schemas.android.c
<EditText
android:layout_height="wrap_content"
android:hint="Enter Your Name" />
<Button
android:id="@+id/btnSave"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_marginTop="15dp"
android:text="Save Name" />
<TextView
android:id="@+id/tvWelcome"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="20dp"
android:text=""
android:textSize="22sp"
android:textStyle="bold" />
<Switch
android:id="@+id/themeSwitch"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="20dp"
android:text="Dark Mode" />
</LinearLayout>
Nisandi
Save Name
Welcome Nisandi
Dark Mode
Practical052 > app > src > main > res > layout > activity_part1.xml
40:16
LF
UTF-8
4 spaces

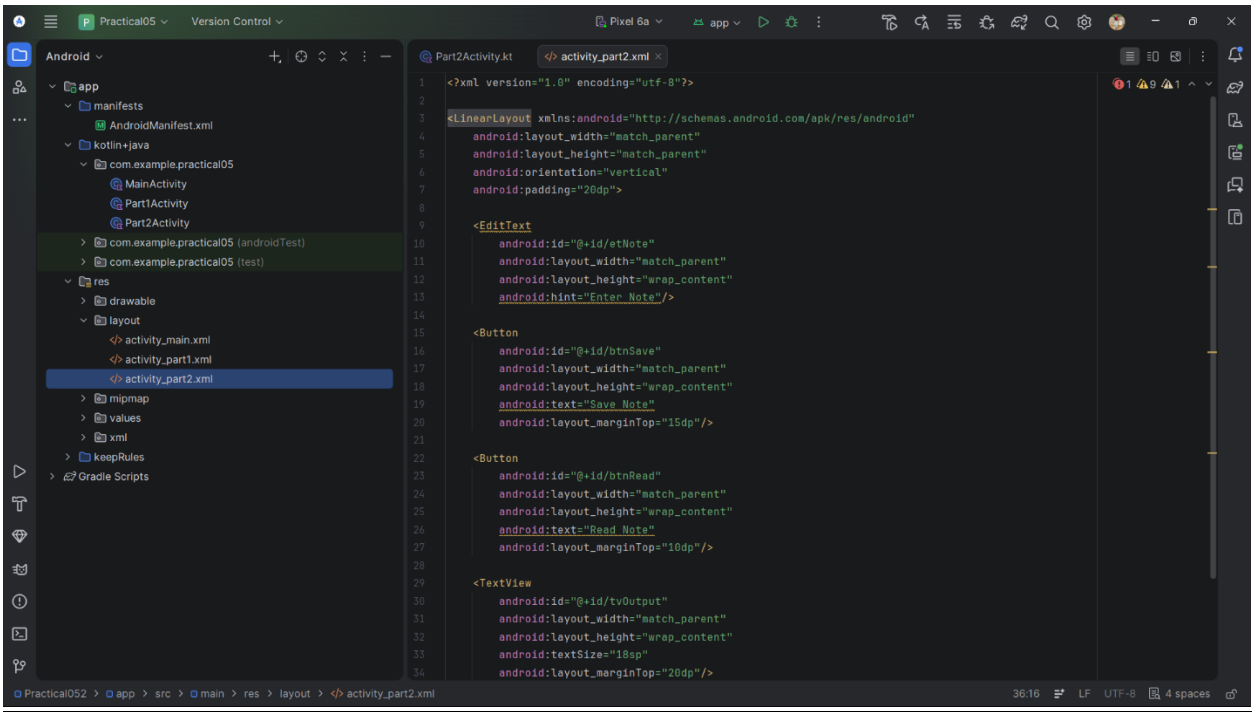


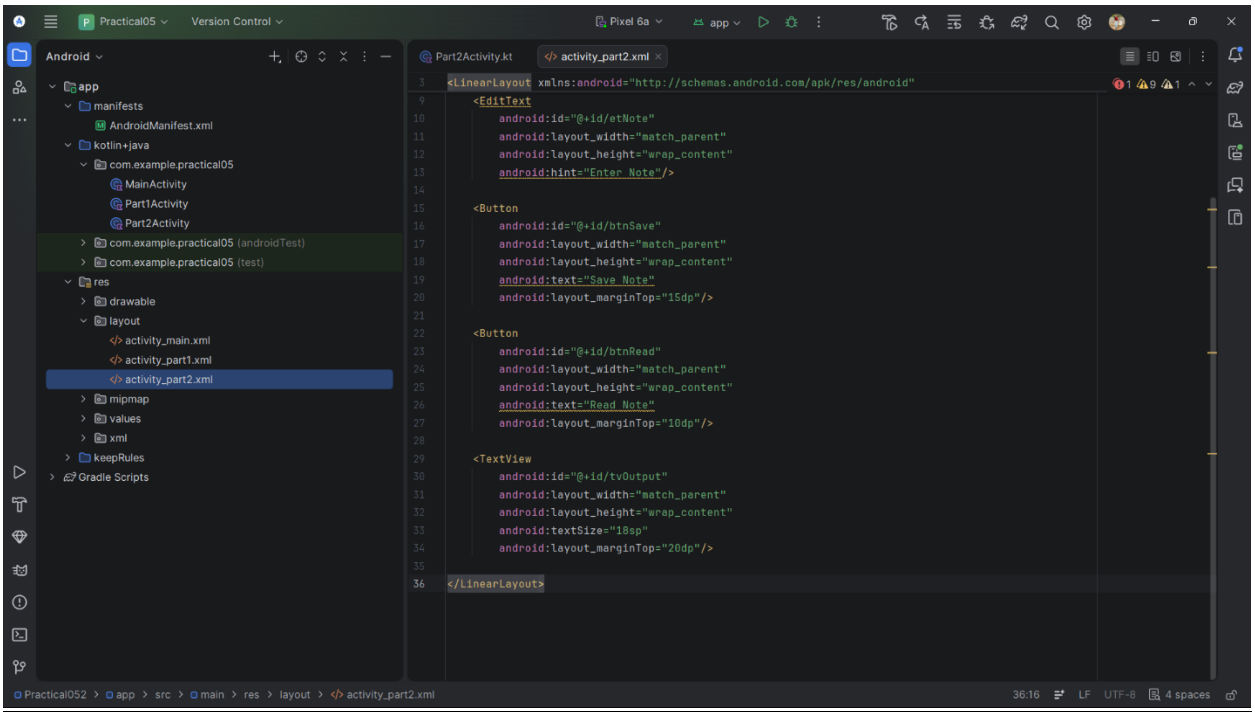
Part 2: App-Specific Storage

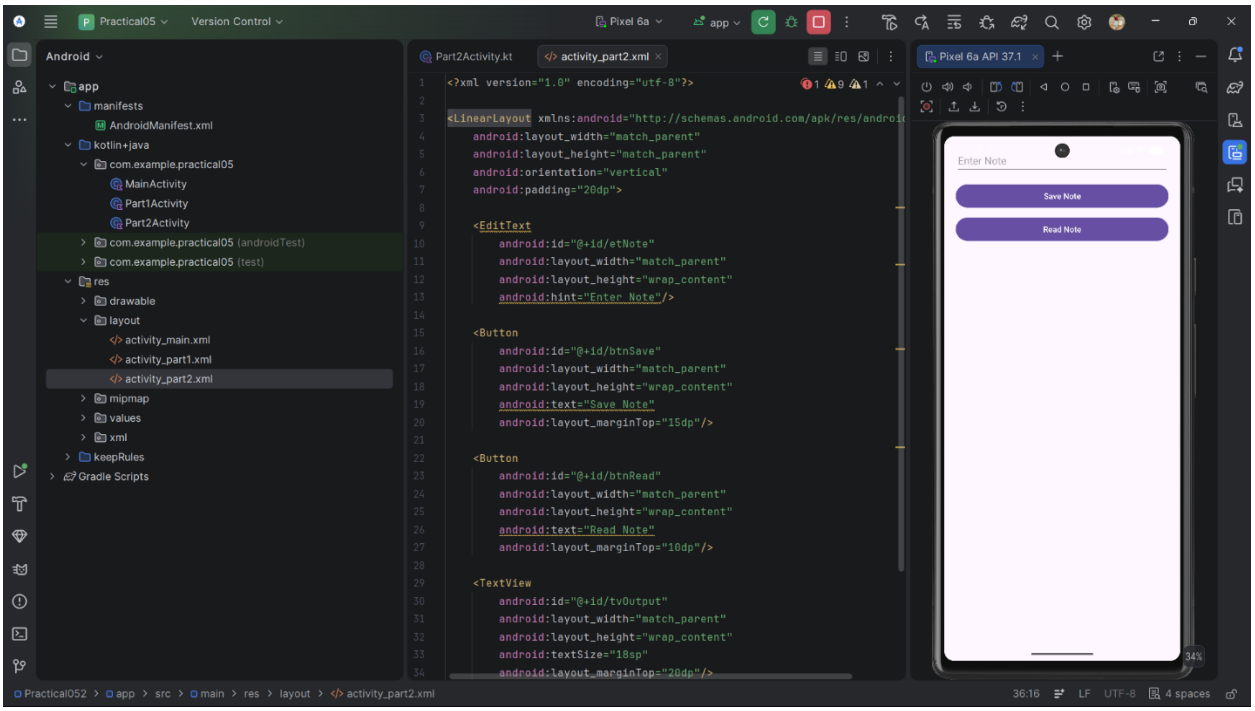


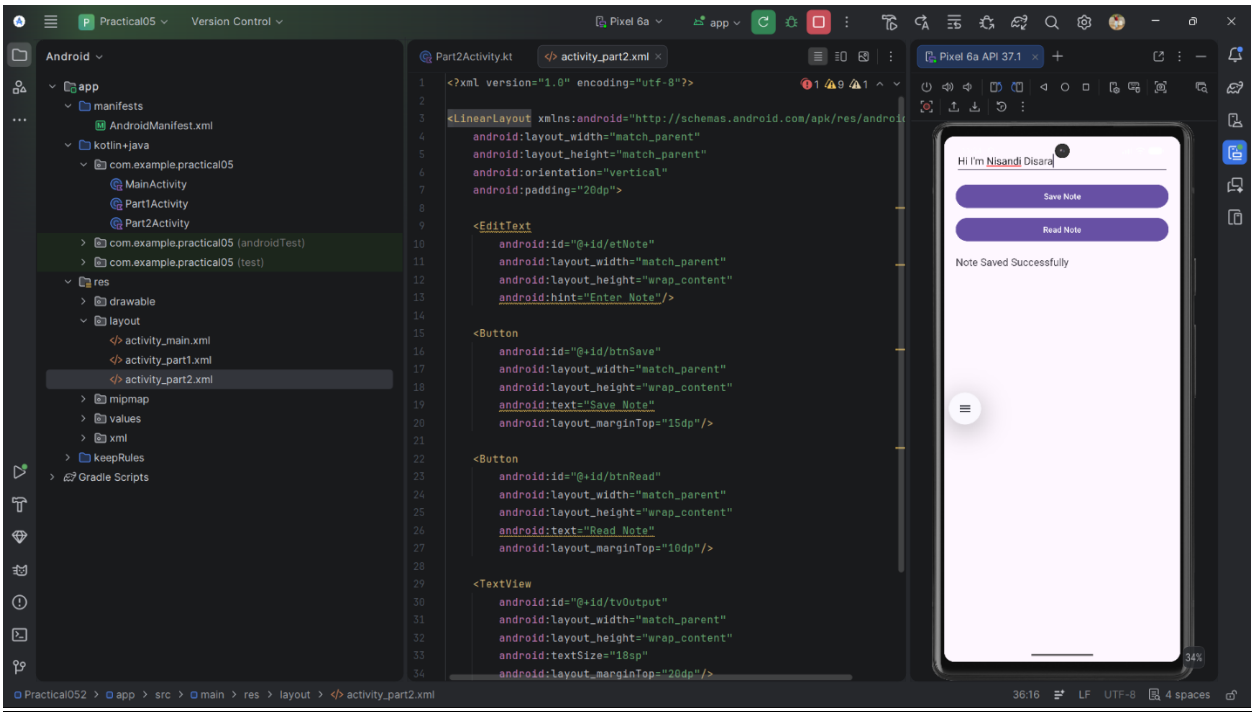

Practical05
Version Control
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app
Android
Part2Activity.kt
app
manifests
AndroidManifest.xml
kotlin+java
com.example.practical05
MainActivity
Part1Activity
Part2Activity
com.example.practical05 (androidTest)
com.example.practical05 (test)
res
drawable
layout
activity_main.xml
activity_part1.xml
activity_part2.xml
mipmap
values
xml
keepRules
Gradle Scripts
class Part2Activity : AppCompatActivity() {
private val cacheFile = "cache.txt"
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_part2)
etNote = findViewById(R.id.etNote)
btnSave = findViewById(R.id.btnSave)
btnRead = findViewById(R.id.btnRead)
tvOutput = findViewById(R.id.tvOutput)
// Check cache on app start
val cache = File(parent = cacheDir, child = cacheFile)
if (cache.exists()) {
tvOutput.text = "Cache : ${cache.readText()}"
}
btnSave.setOnClickListener {
val note = etNote.text.toString()
// Save in filesDir
val file = File(parent = filesDir, child = fileName)
file.writeText(note)
// Save in cacheDir
val cacheFileObj = File(parent = cacheDir, child = cacheFile)
cacheFileObj.writeText(note)
tvOutput.text = "Note Saved Successfully"
}
Practical052 › app › src › main › java › com › example › practical05 › Part2Activity
62:2
LF
UTF-8
4 spaces

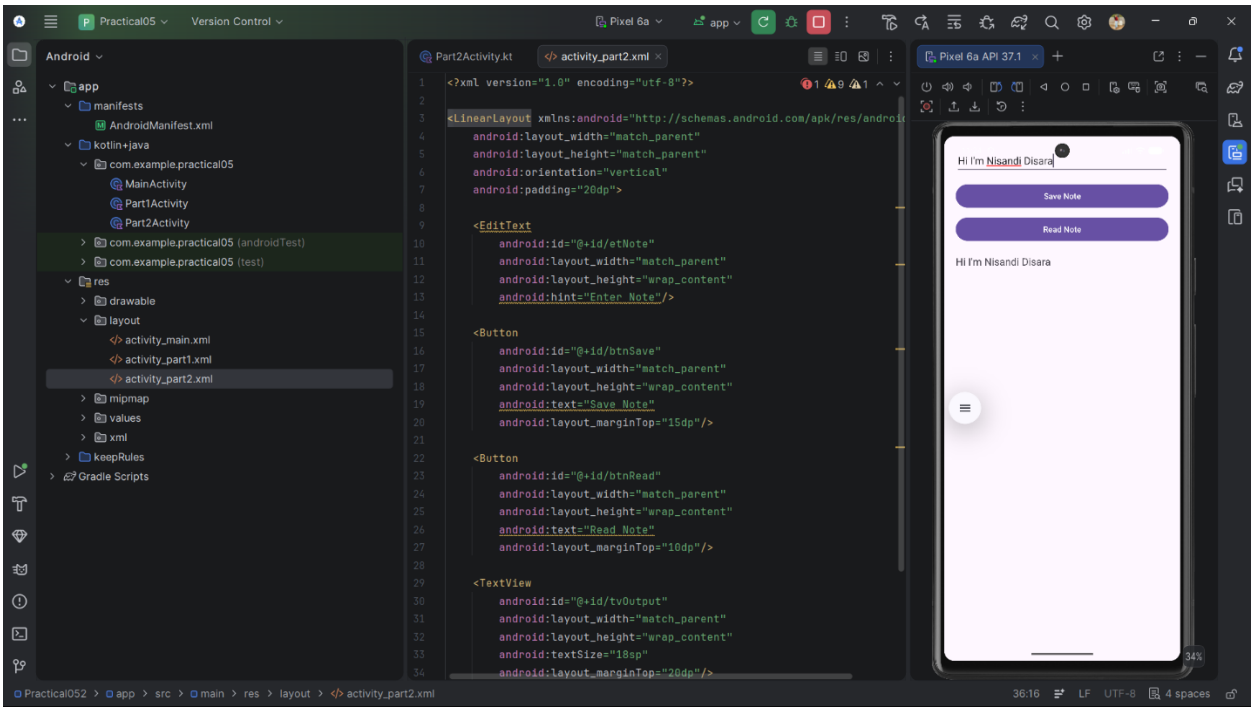

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Android
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com.example.practical05
MainActivity
Part1Activity
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com.example.practical05 (androidTest)
com.example.practical05 (test)
res
drawable
layout
activity_main.xml
activity_part1.xml
activity_part2.xml
mipmap
values
xml
keepRules
Gradle Scripts
class Part2Activity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
tvOutput.text = "Cache : ${cache.readText()}"
}
btnSave.setOnClickListener {
val note = etNote.text.toString()
// Save in filesDir
val file = File(parent = filesDir, child = fileName)
file.writeText(note)
// Save in cacheDir
val cacheFileObj = File(parent = cacheDir, child = cacheFile)
cacheFileObj.writeText(note)
tvOutput.text = "Note Saved Successfully"
}
btnRead.setOnClickListener {
val file = File(parent = filesDir, child = fileName)
if (file.exists()) {
tvOutput.text = file.readText()
} else {
tvOutput.text = "No Note Found"
}
}
}
}
Practical052 › app › src › main › java › com › example › practical05 › Part2Activity
62:2
LF
UTF-8
4 spaces


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Android
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MainActivity
Part1Activity
Part2Activity
com.example.practical05 (androidTest)
com.example.practical05 (test)
res
drawable
layout
activity_main.xml
activity_part1.xml
activity_part2.xml
mipmap
values
xml
keepRules
Gradle Scripts
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp">
<EditText
android:id="@+id/etNote"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter Note"/>
<Button
android:id="@+id/btnSave"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Save Note"
android:layout_marginTop="15dp"/>
<Button
android:id="@+id/btnRead"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Read Note"
android:layout_marginTop="10dp"/>
<TextView
android:id="@+id/tvOutput"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="18sp"
android:layout_marginTop="20dp"/>
Practical052 › app › src › main › res › layout › activity_part2.xml
36:16
LF
UTF-8
4 spaces

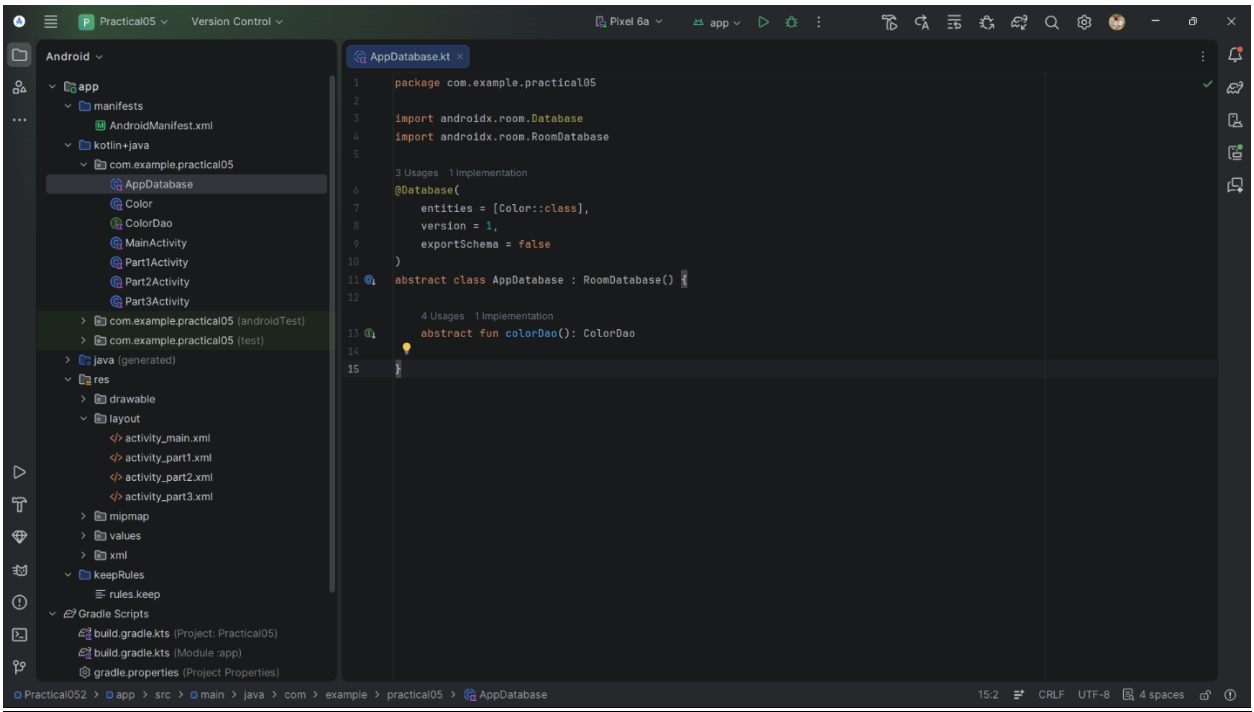

Practical05
Version Control
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Android
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AndroidManifest.xml
kotlin+java
com.example.practical05
MainActivity
Part1Activity
Part2Activity
com.example.practical05 (androidTest)
com.example.practical05 (test)
res
drawable
layout
activity_main.xml
activity_part1.xml
activity_part2.xml
mipmap
values
xml
keepRules
Gradle Scripts
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
<EditText
android:id="@+id/etNote"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter Note"/>
<Button
android:id="@+id/btnSave"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Save Note"
android:layout_marginTop="15dp"/>
<Button
android:id="@+id/btnRead"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Read Note"
android:layout_marginTop="10dp"/>
<TextView
android:id="@+id/tvOutput"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="18sp"
android:layout_marginTop="20dp"/>
</LinearLayout>
Practical052 › app › src › main › res › layout › activity_part2.xml
36:16
LF
UTF-8
4 spaces

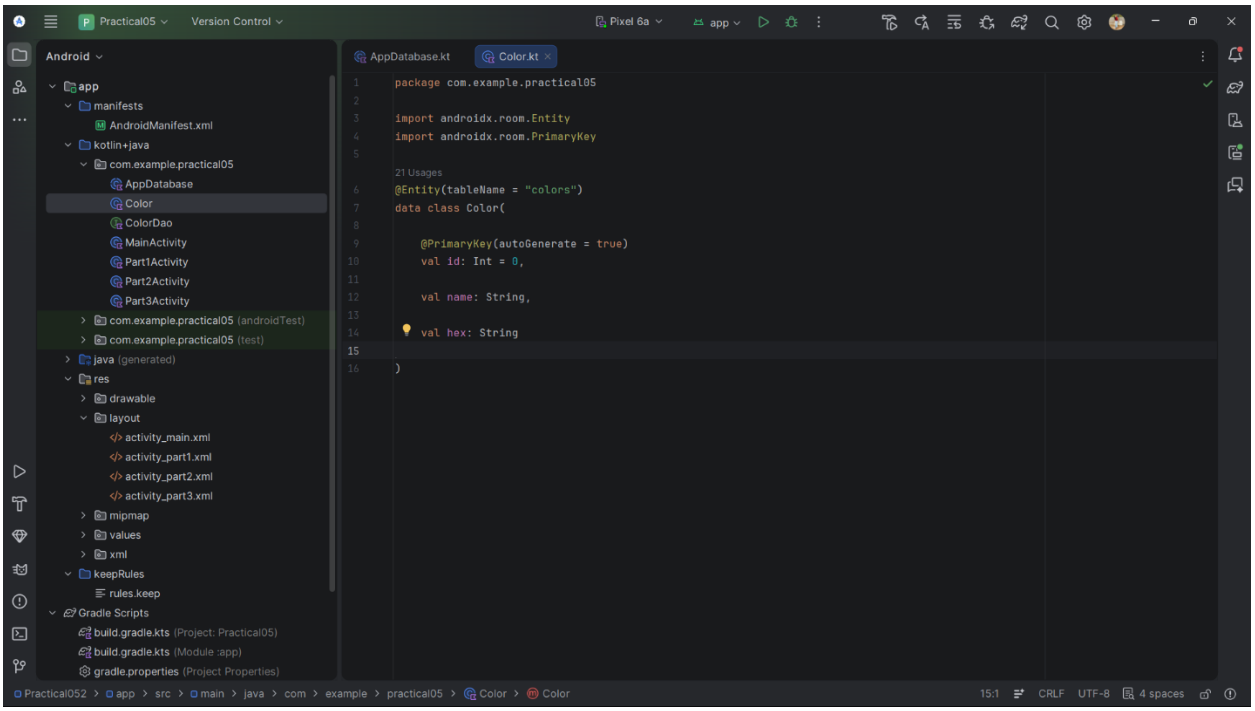

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MainActivity
Part1Activity
Part2Activity
com.example.practical05 (androidTest)
com.example.practical05 (test)
res
drawable
layout
activity_main.xml
activity_part1.xml
activity_part2.xml
mipmap
values
xml
keepRules
Gradle Scripts
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/androi
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp">
<EditText
android:id="@+id/etNote"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter Note"/>
<Button
android:id="@+id/btnSave"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Save Note"
android:layout_marginTop="15dp"/>
<Button
android:id="@+id/btnRead"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Read Note"
android:layout_marginTop="10dp"/>
<TextView
android:id="@+id/tvOutput"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="18sp"
android:layout_marginTop="20dp"/>
Enter Note
Save Note
Read Note
34%
Practical052 › app › src › main › res › layout › activity_part2.xml
36:16
LF
UTF-8
4 spaces

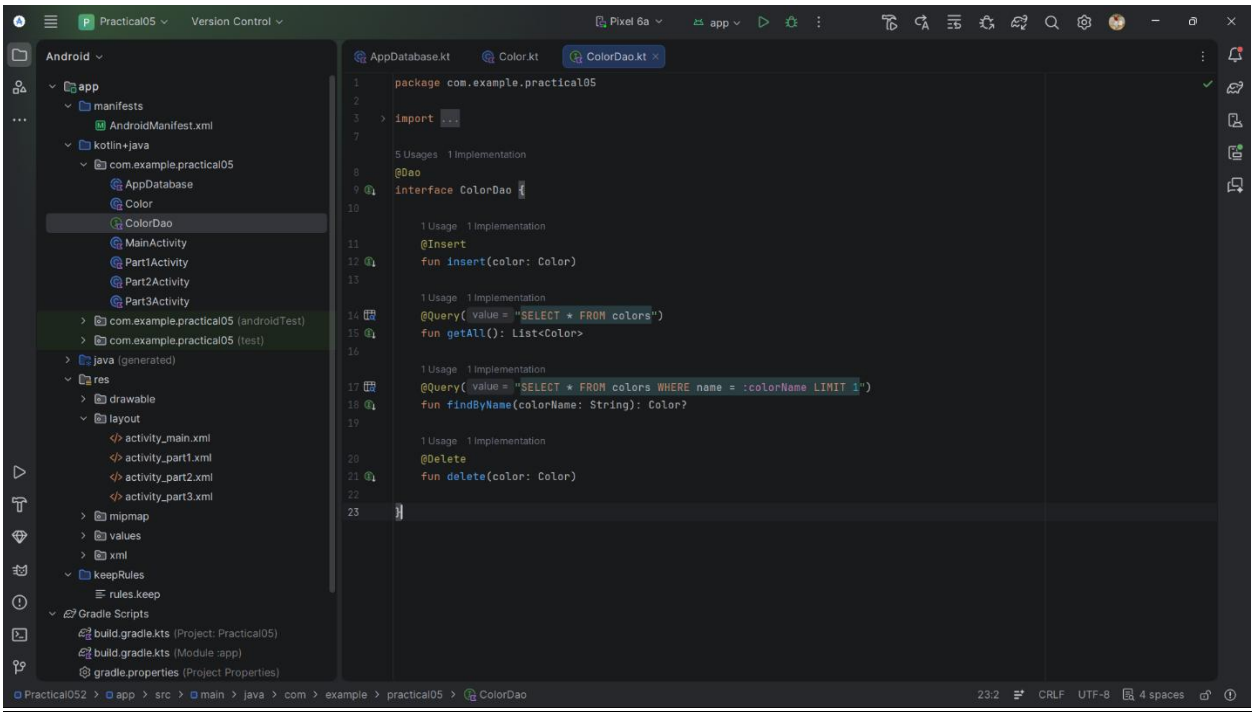

Practical05
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Pixel 6a
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Android
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activity_part2.xml
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com.example.practical05
MainActivity
Part1Activity
Part2Activity
com.example.practical05 (androidTest)
com.example.practical05 (test)
res
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activity_main.xml
activity_part1.xml
activity_part2.xml
mipmap
values
xml
keepRules
Gradle Scripts
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/androi
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp">
<EditText
android:id="@+id/etNote"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter Note"/>
<Button
android:id="@+id/btnSave"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Save Note"
android:layout_marginTop="15dp"/>
<Button
android:id="@+id/btnRead"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Read Note"
android:layout_marginTop="10dp"/>
<TextView
android:id="@+id/tvOutput"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="18sp"
android:layout_marginTop="20dp"/>
Hi I'm Nisandi Disara
Save Note
Read Note
Note Saved Successfully
34%
Practical052 › app › src › main › res › layout › activity_part2.xml
36:16
LF
UTF-8
4 spaces

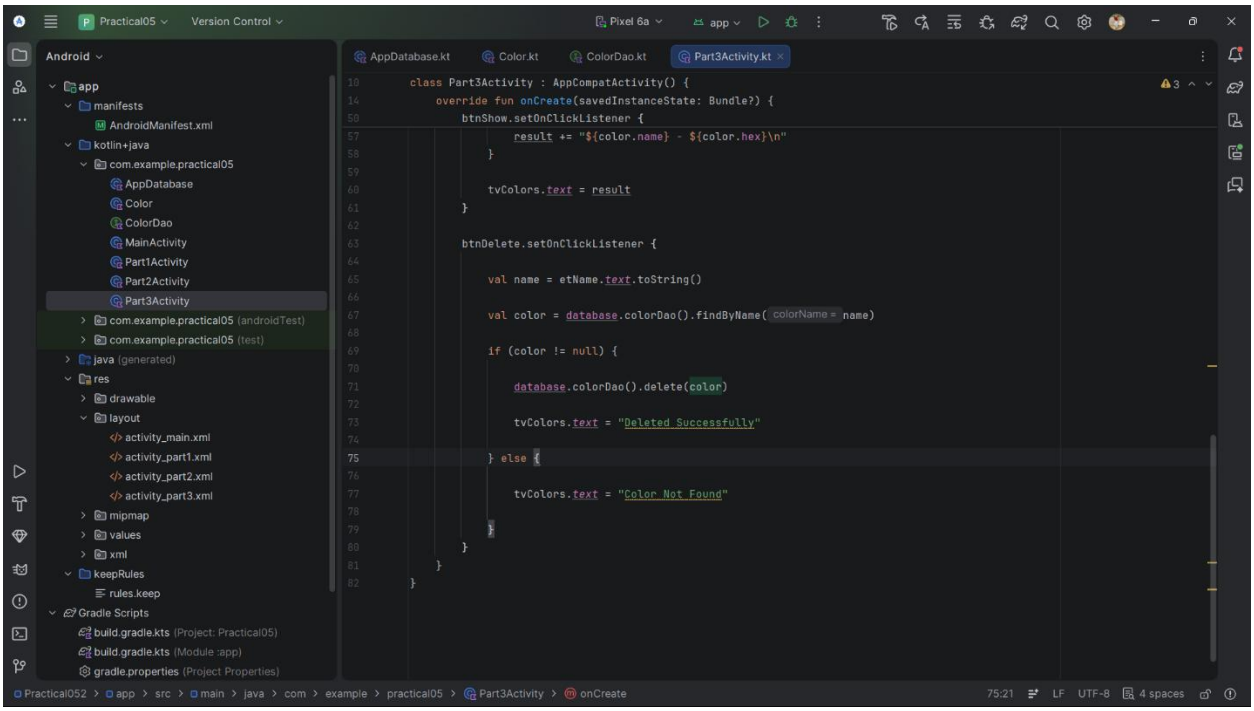

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com.example.practical05
MainActivity
Part1Activity
Part2Activity
com.example.practical05 (androidTest)
com.example.practical05 (test)
res
drawable
layout
activity_main.xml
activity_part1.xml
activity_part2.xml
mipmap
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keepRules
Gradle Scripts
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp">
<EditText
android:id="@+id/etNote"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter Note"/>
<Button
android:id="@+id/btnSave"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Save Note"
android:layout_marginTop="15dp"/>
<Button
android:id="@+id/btnRead"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Read Note"
android:layout_marginTop="10dp"/>
<TextView
android:id="@+id/tvOutput"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="18sp"
android:layout_marginTop="20dp"/>
Hi I'm Nisandi Disara
Save Note
Read Note
Hi I'm Nisandi Disara
Practical052 > app > src > main > res > layout > activity_part2.xml
36:16
LF
UTF-8
4 spaces

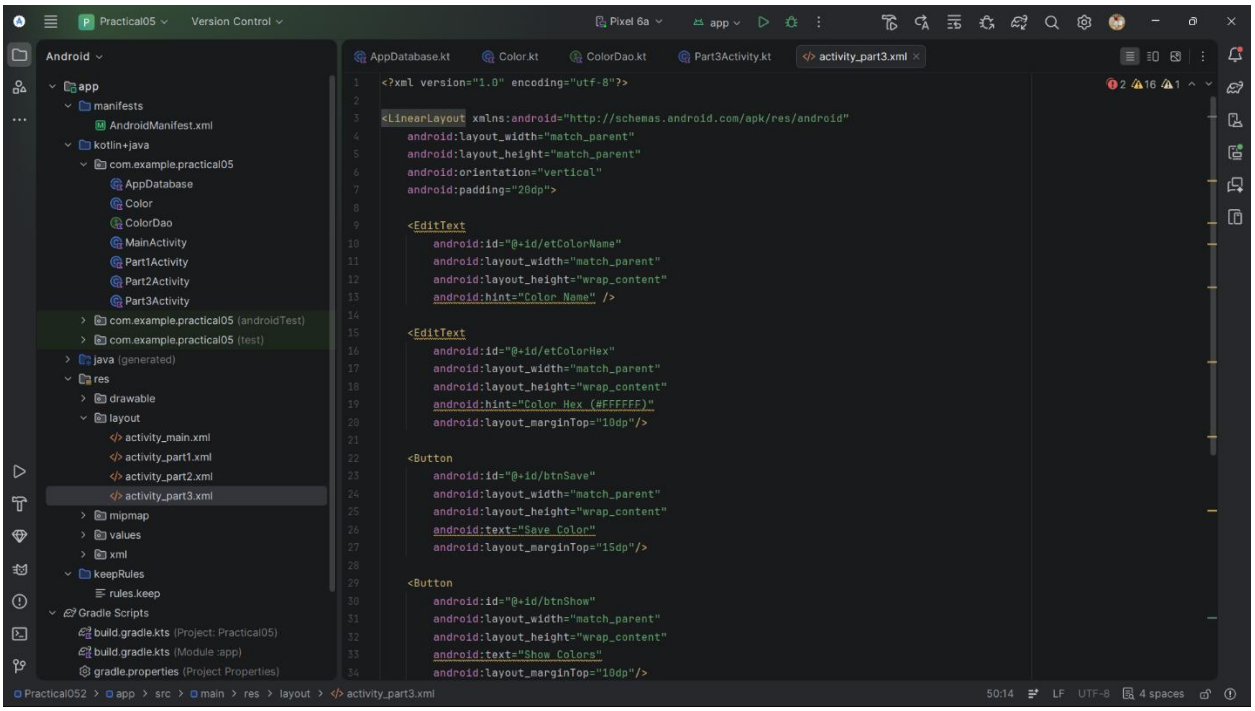
Part 3: Room

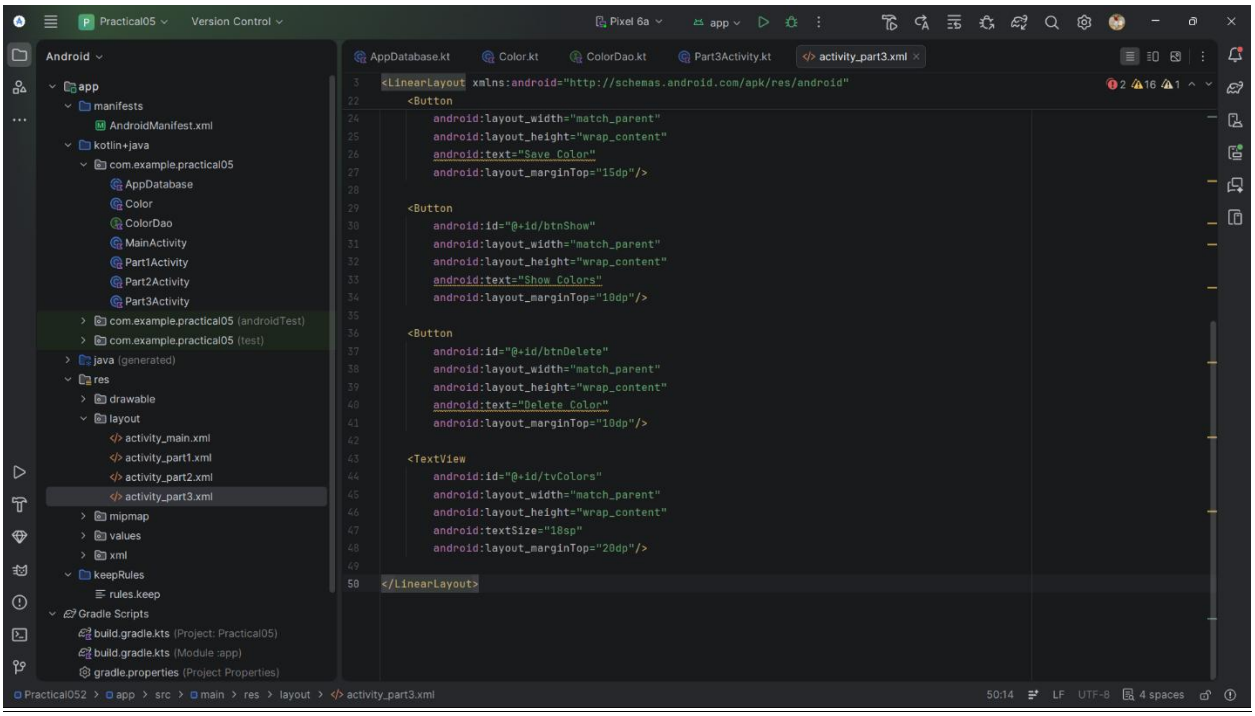

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com.example.practical05 (test)
java (generated)
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activity_part1.xml
activity_part2.xml
activity_part3.xml
mipmap
values
xml
keepRules
rules.keep
Gradle Scripts
build.gradle.kts (Project: Practical05)
build.gradle.kts (Module :app)
gradle.properties (Project Properties)
package com.example.practical05
import androidx.room.Database
import androidx.room.RoomDatabase
3 Usages 1 Implementation
@Database(
entities = [Color::class],
version = 1,
exportSchema = false
)
abstract class AppDatabase : RoomDatabase() {
4 Usages 1 Implementation
abstract fun colorDao(): ColorDao
}
Practical052 > app > src > main > java > com > example > practical05 > AppDatabase
15:2
CRLF
UTF-8
4 spaces

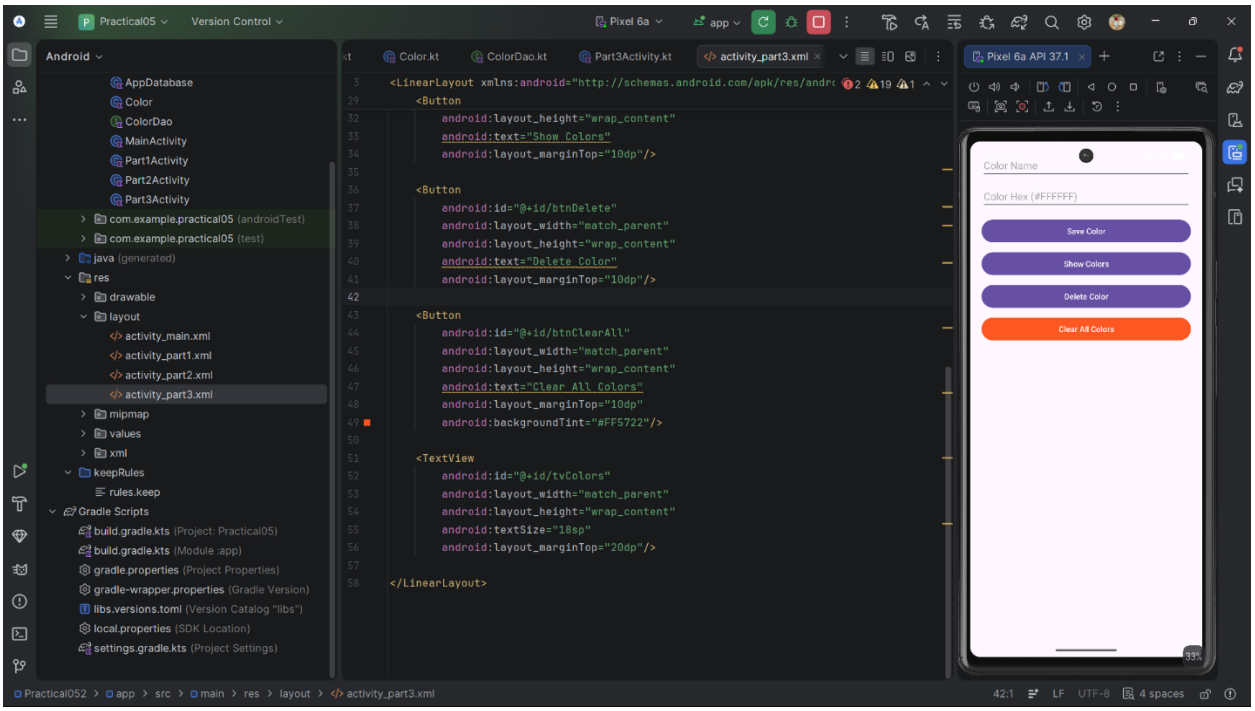

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Gradle Scripts
build.gradle.kts (Project: Practical05)
build.gradle.kts (Module :app)
gradle.properties (Project Properties)
package com.example.practical05
import androidx.room.Entity
import androidx.room.PrimaryKey
21 Usages
@Entity(tableName = "colors")
data class Color(
@PrimaryKey(autoGenerate = true)
val id: Int = 0,
val name: String,
val hex: String
)
Practical052 › app › src › main › java › com › example › practical05 › Color › Color
15:1
CRLF
UTF-8
4 spaces

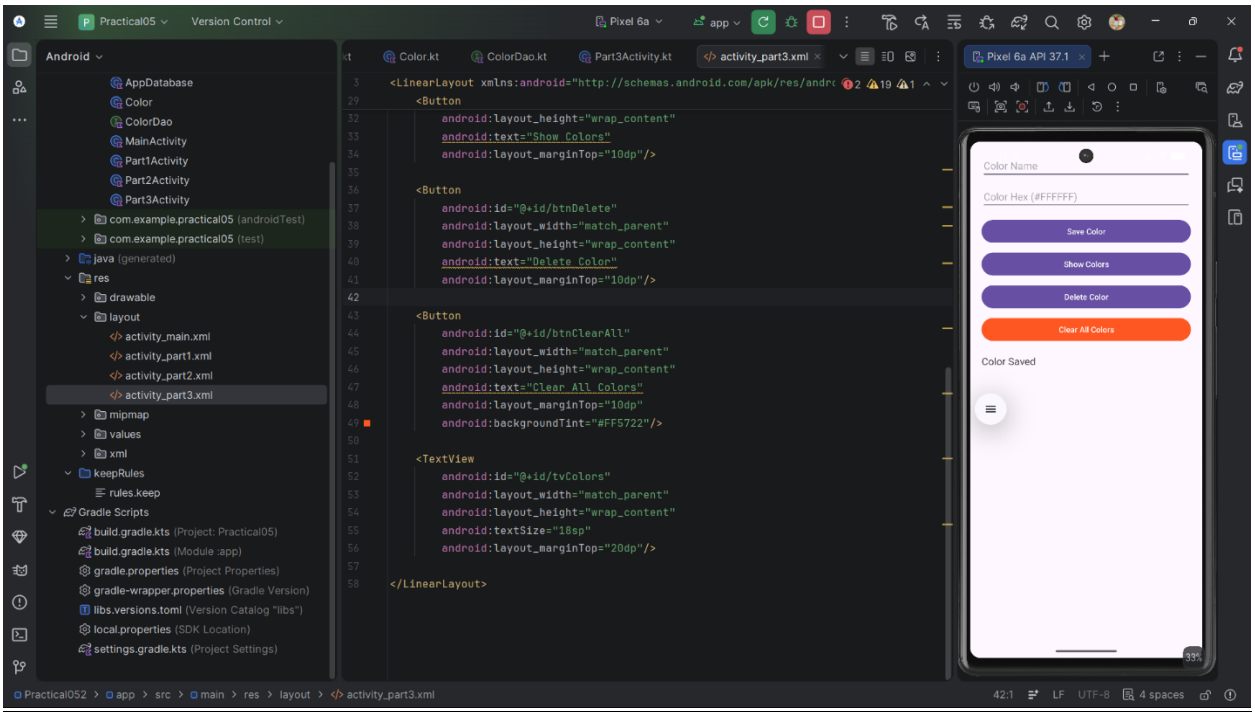

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Gradle Scripts
build.gradle.kts (Project: Practical05)
build.gradle.kts (Module :app)
gradle.properties (Project Properties)
package com.example.practical05
import ...
5 Usages 1 Implementation
@Dao
interface ColorDao {
1 Usage 1 Implementation
@Insert
fun insert(color: Color)
1 Usage 1 Implementation
@Query(value = "SELECT * FROM colors")
fun getAll(): List<Color>
1 Usage 1 Implementation
@Query(value = "SELECT * FROM colors WHERE name = :colorName LIMIT 1")
fun findByName(colorName: String): Color?
1 Usage 1 Implementation
@Delete
fun delete(color: Color)
}
Practical052 › app › src › main › java › com › example › practical05 › ColorDao
23:2
CRLF
UTF-8
4 spaces

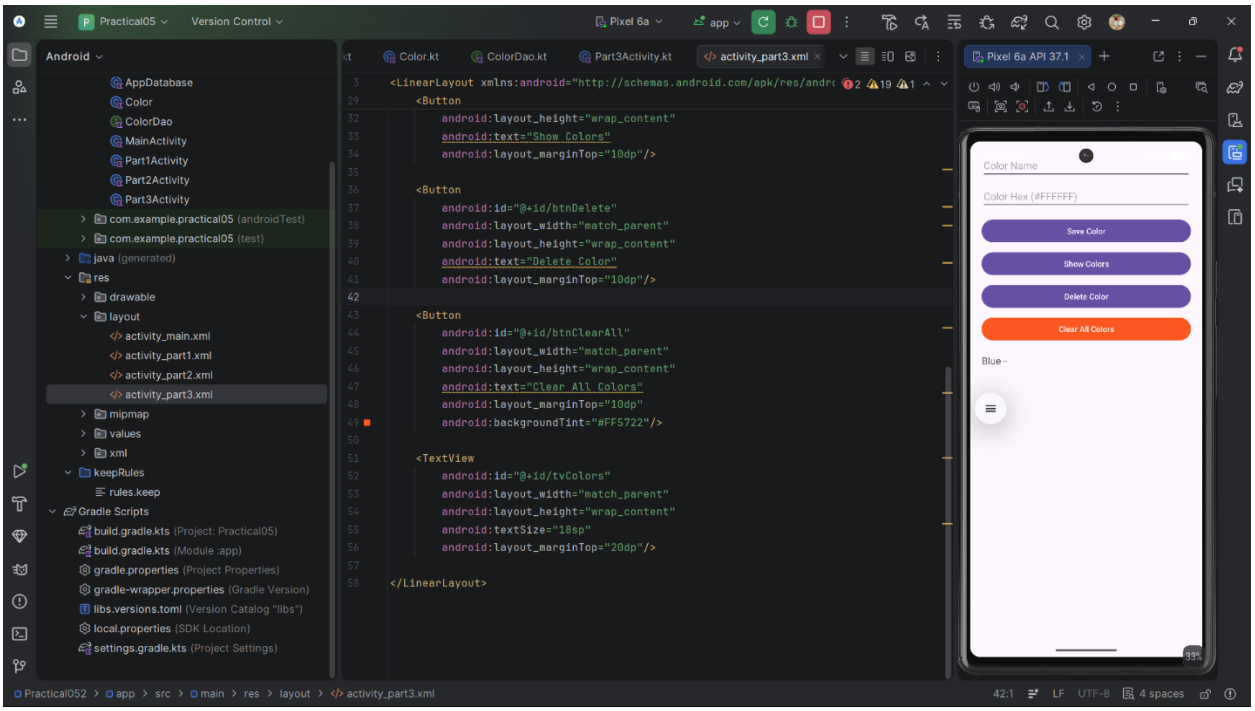

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com.example.practical05 (test)
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AppDatabase.kt
Color.kt
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Part3Activity.kt
class Part3Activity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
btnShow.setOnClickListener {
result += "${color.name} - ${color.hex}\n"
}
tvColors.text = result
}
btnDelete.setOnClickListener {
val name = etName.text.toString()
val color = database.colorDao().findByName(colorName = name)
if (color != null) {
database.colorDao().delete(color)
tvColors.text = "Deleted Successfully"
} else {
tvColors.text = "Color Not Found"
}
}
}
}
Practical052 › app › src › main › java › com › example › practical05 › Part3Activity › onCreate
75:21
LF
UTF-8
4 spaces

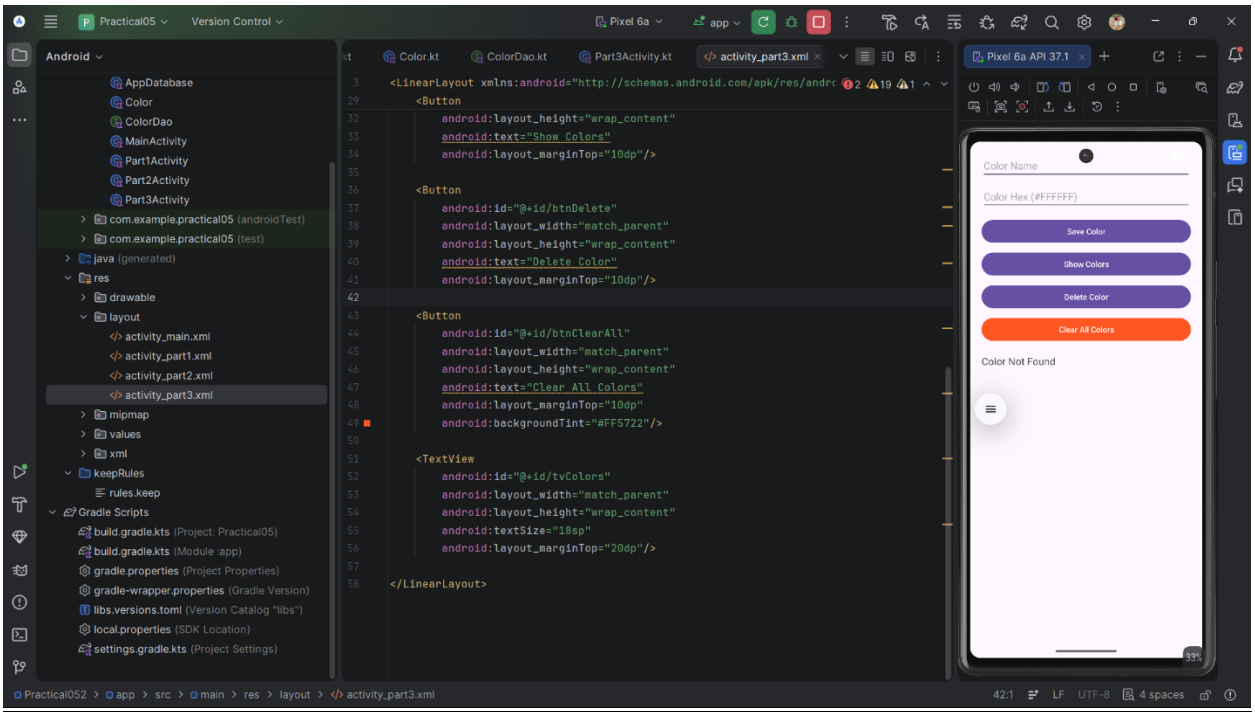

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build.gradle.kts (Project: Practical05)
build.gradle.kts (Module :app)
gradle.properties (Project Properties)
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp">
<EditText
android:id="@+id/etColorName"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Color Name" />
<EditText
android:id="@+id/etColorHex"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Color Hex (#FFFFFF)"
android:layout_marginTop="10dp"/>
<Button
android:id="@+id/btnSave"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Save Color"
android:layout_marginTop="15dp"/>
<Button
android:id="@+id/btnShow"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Show Colors"
android:layout_marginTop="10dp"/>
Practical052 > app > src > main > res > layout > activity_part3.xml
50:14
LF
UTF-8
4 spaces

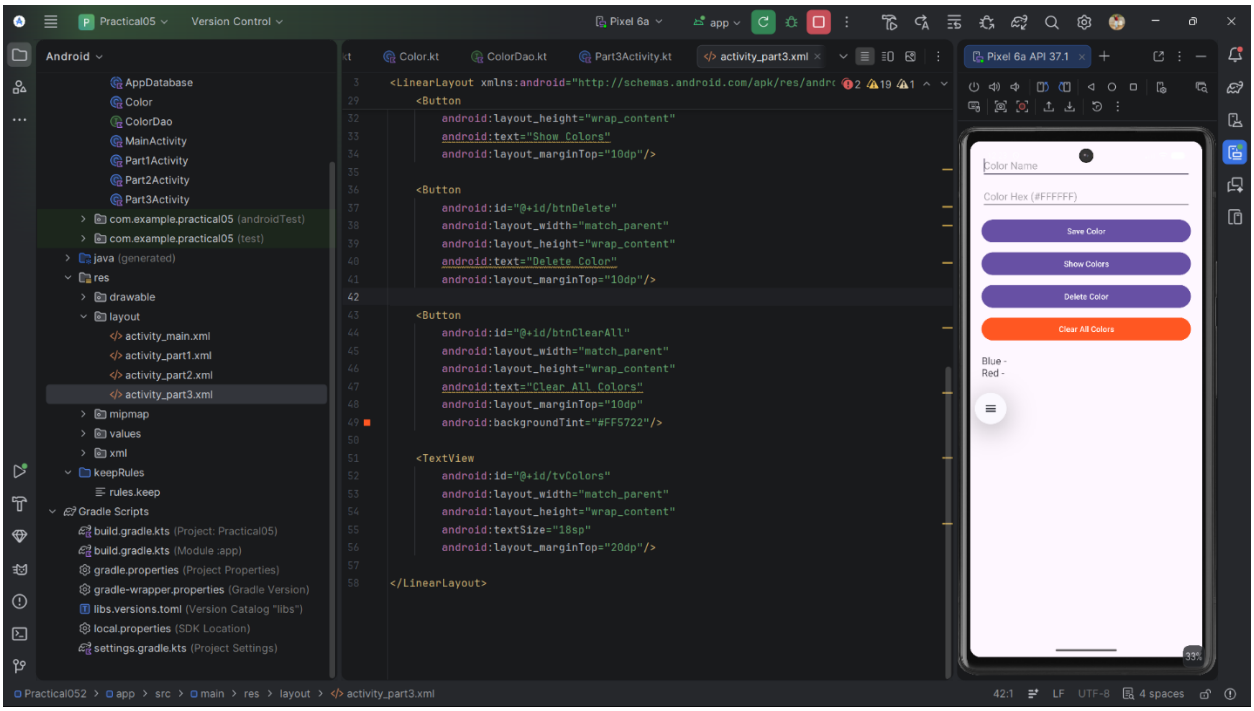

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<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
<Button
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Save Color"
android:layout_marginTop="15dp"/>
<Button
android:id="@+id/btnShow"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Show Colors"
android:layout_marginTop="10dp"/>
<Button
android:id="@+id/btnDelete"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Delete Color"
android:layout_marginTop="10dp"/>
<TextView
android:id="@+id/tvColors"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="18sp"
android:layout_marginTop="20dp"/>
</LinearLayout>
Practical052 > app > src > main > res > layout > activity_part3.xml
50:14
LF
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local.properties (SDK Location)
settings.gradle.kts (Project Settings)
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activity_part3.xml
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/andr
<Button
android:layout_height="wrap_content"
android:text="Show Colors"
android:layout_marginTop="10dp"/>
<Button
android:id="@+id/btnDelete"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Delete Color"
android:layout_marginTop="10dp"/>
<Button
android:id="@+id/btnClearAll"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Clear All Colors"
android:layout_marginTop="10dp"
android:backgroundTint="#FF5722"/>
<TextView
android:id="@+id/tvColors"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="18sp"
android:layout_marginTop="20dp"/>
</LinearLayout>
Pixel 6a API 37.1
Color Name
Color Hex (#FFFFFF)
Save Color
Show Colors
Delete Color
Clear All Colors
Practical052 › app › src › main › res › layout › activity_part3.xml
42:1
LF
UTF-8
4 spaces

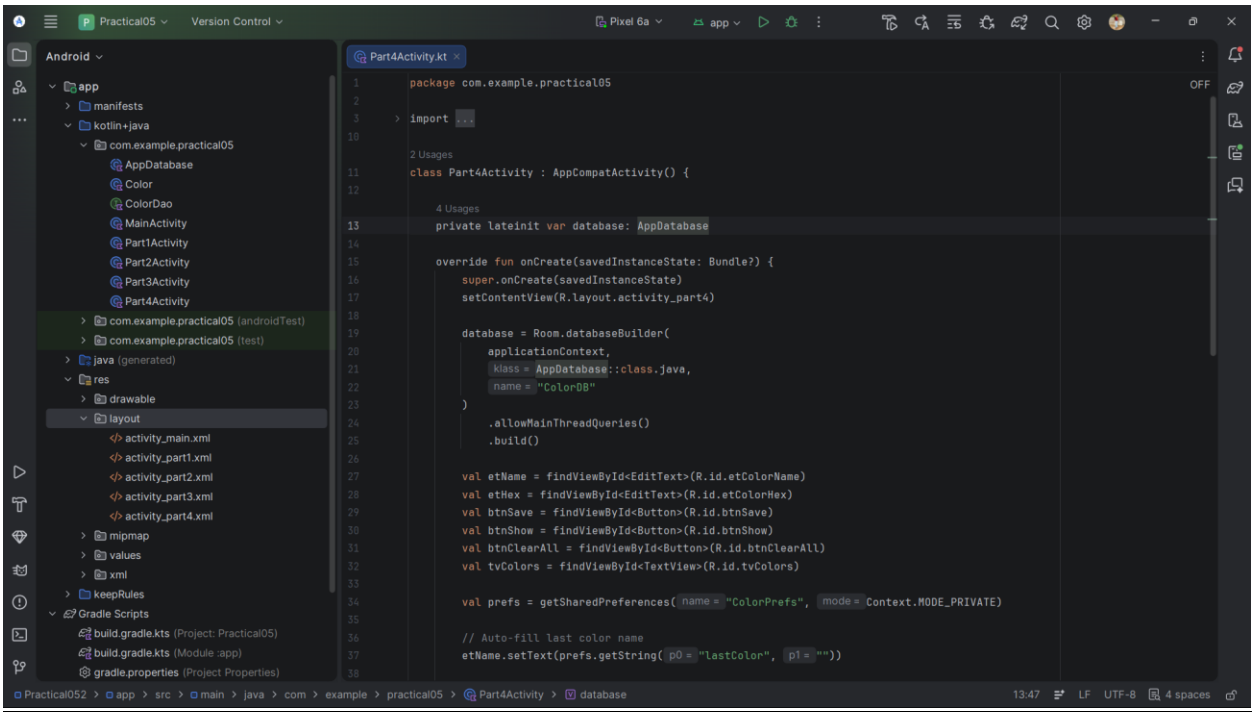

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libs.versions.toml (Version Catalog "libs")
local.properties (SDK Location)
settings.gradle.kts (Project Settings)
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activity_part3.xml
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/andr
<Button
android:layout_height="wrap_content"
android:text="Show Colors"
android:layout_marginTop="10dp"/>
<Button
android:id="@+id/btnDelete"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Delete Color"
android:layout_marginTop="10dp"/>
<Button
android:id="@+id/btnClearAll"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Clear All Colors"
android:layout_marginTop="10dp"
android:backgroundTint="#FF5722"/>
<TextView
android:id="@+id/tvColors"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="18sp"
android:layout_marginTop="20dp"/>
</LinearLayout>
Pixel 6a API 37.1
Color Name
Color Hex (#FFFFFF)
Save Color
Show Colors
Delete Color
Clear All Colors
Color Saved
Practical052 › app › src › main › res › layout › activity_part3.xml
42:1
LF
UTF-8
4 spaces

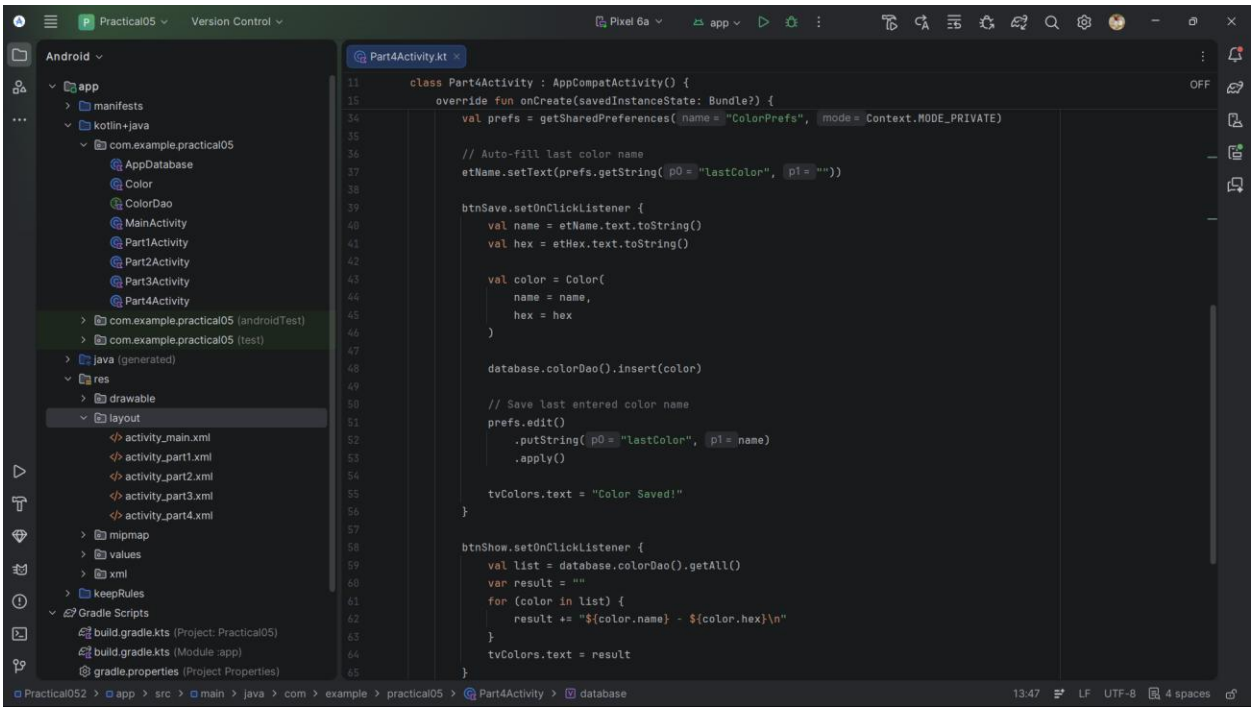

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com.example.practical05 (test)
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build.gradle.kts (Module :app)
gradle.properties (Project Properties)
gradle-wrapper.properties (Gradle Version)
libs.versions.toml (Version Catalog "libs")
local.properties (SDK Location)
settings.gradle.kts (Project Settings)
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/andr
<Button
android:layout_height="wrap_content"
android:text="Show Colors"
android:layout_marginTop="10dp"/>
<Button
android:id="@+id/btnDelete"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Delete Color"
android:layout_marginTop="10dp"/>
<Button
android:id="@+id/btnClearAll"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Clear All Colors"
android:layout_marginTop="10dp"
android:backgroundTint="#FF5722"/>
<TextView
android:id="@+id/tvColors"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="18sp"
android:layout_marginTop="20dp"/>
</LinearLayout>
Color Name
Color Hex (#FFFFFF)
Save Color
Show Colors
Delete Color
Clear All Colors
Blue -
Practical052 › app › src › main › res › layout › activity_part3.xml
42:1
LF
UTF-8
4 spaces

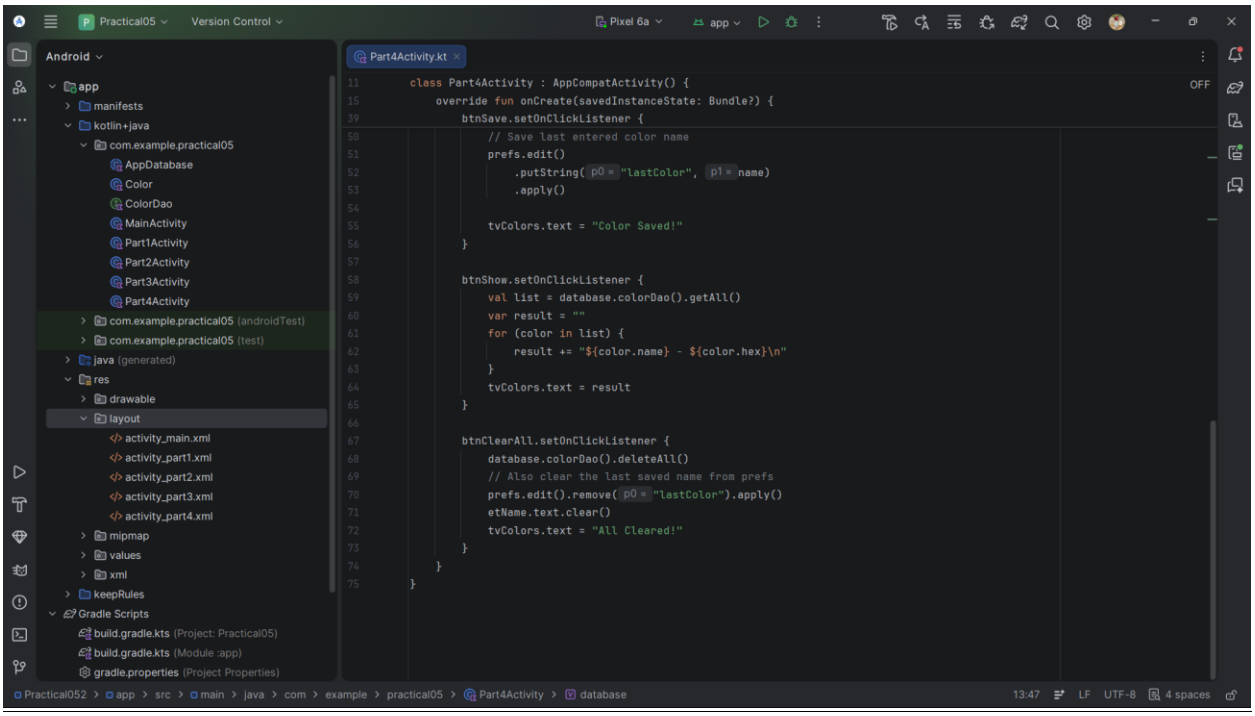

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libs.versions.toml (Version Catalog "libs")
local.properties (SDK Location)
settings.gradle.kts (Project Settings)
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/andr
<Button
android:layout_height="wrap_content"
android:text="Show Colors"
android:layout_marginTop="10dp"/>
<Button
android:id="@+id/btnDelete"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Delete Color"
android:layout_marginTop="10dp"/>
<Button
android:id="@+id/btnClearAll"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Clear All Colors"
android:layout_marginTop="10dp"
android:backgroundTint="#FF5722"/>
<TextView
android:id="@+id/tvColors"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="18sp"
android:layout_marginTop="20dp"/>
</LinearLayout>
Color Name
Color Hex (#FFFFFF)
Save Color
Show Colors
Delete Color
Clear All Colors
Color Not Found
Practical052 › app › src › main › res › layout › activity_part3.xml
42:1
LF
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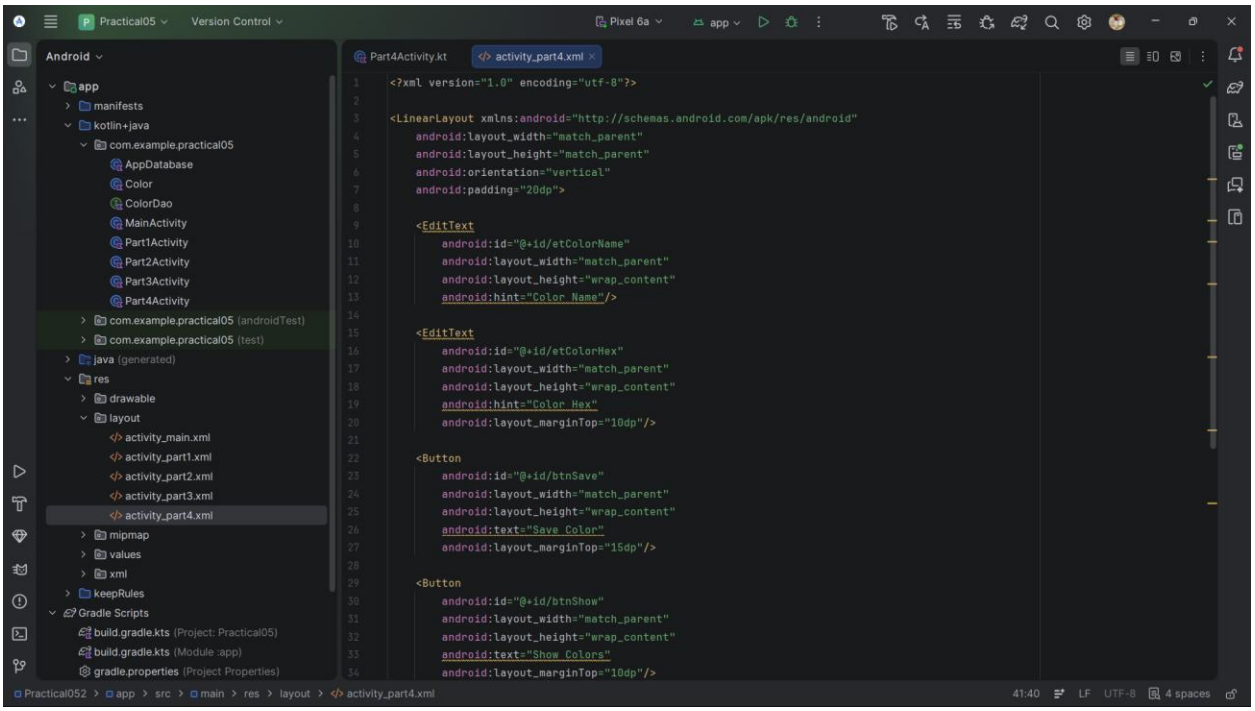


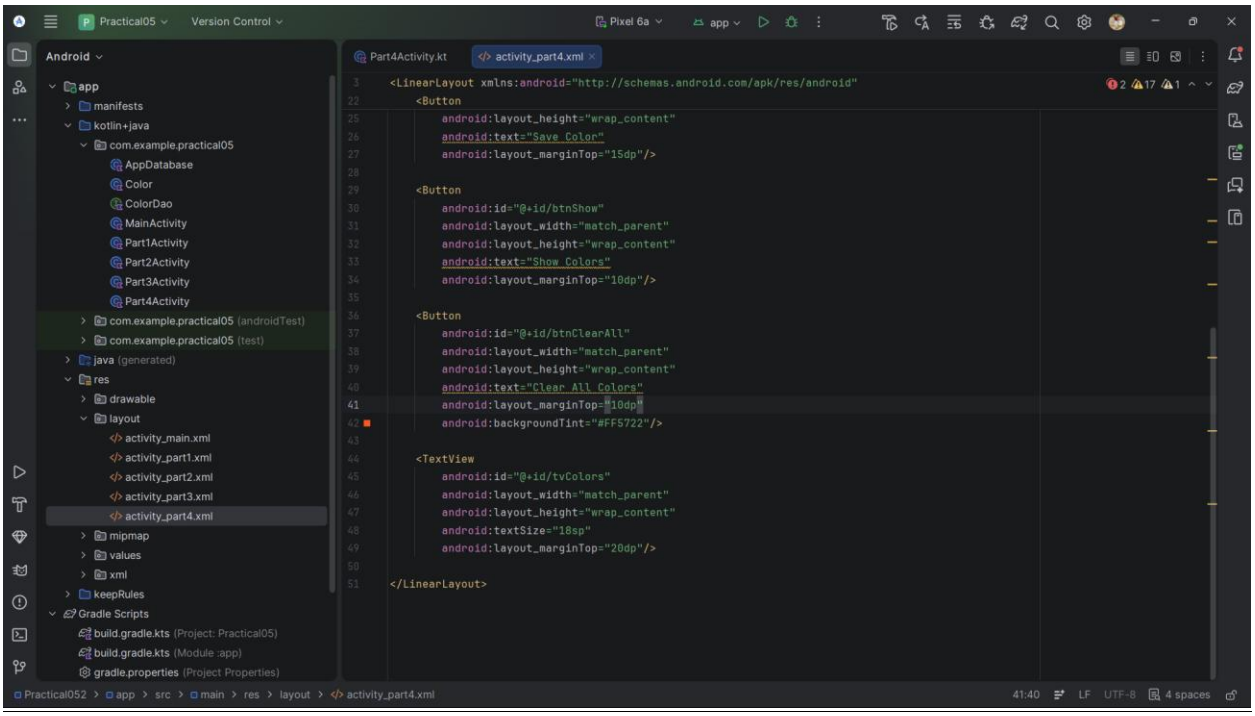
Part 4: Room + SharedPreferences Combined

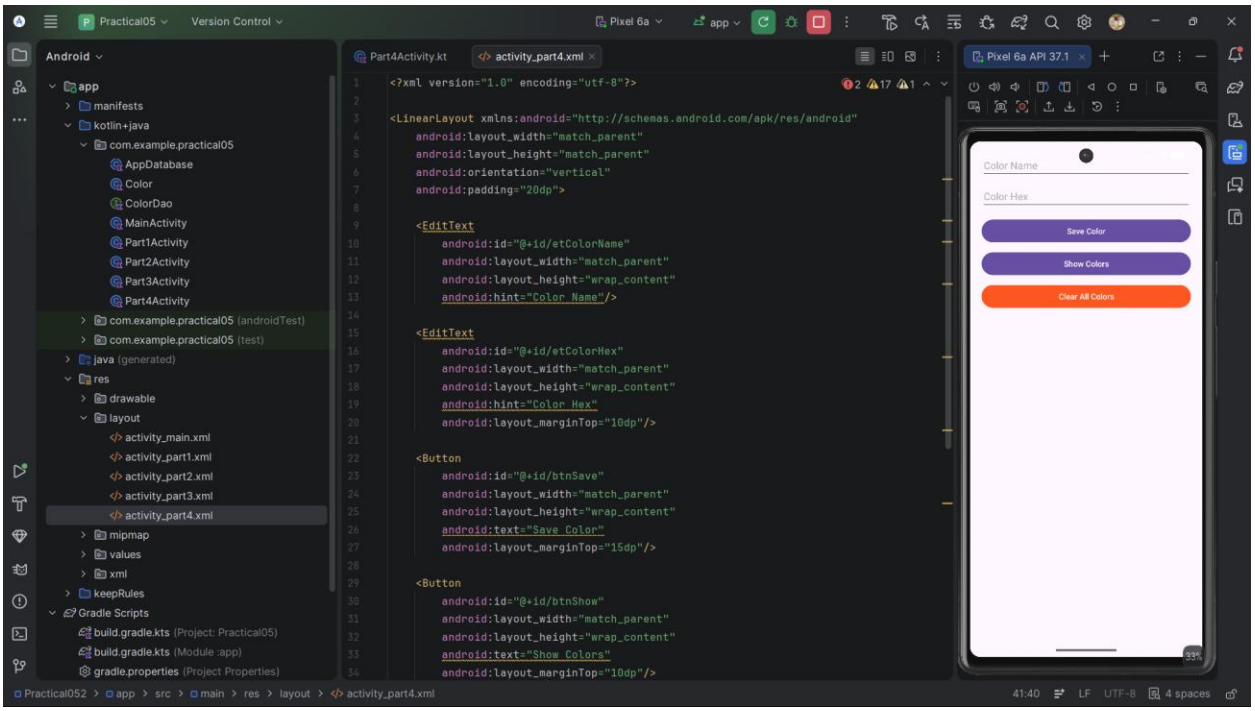


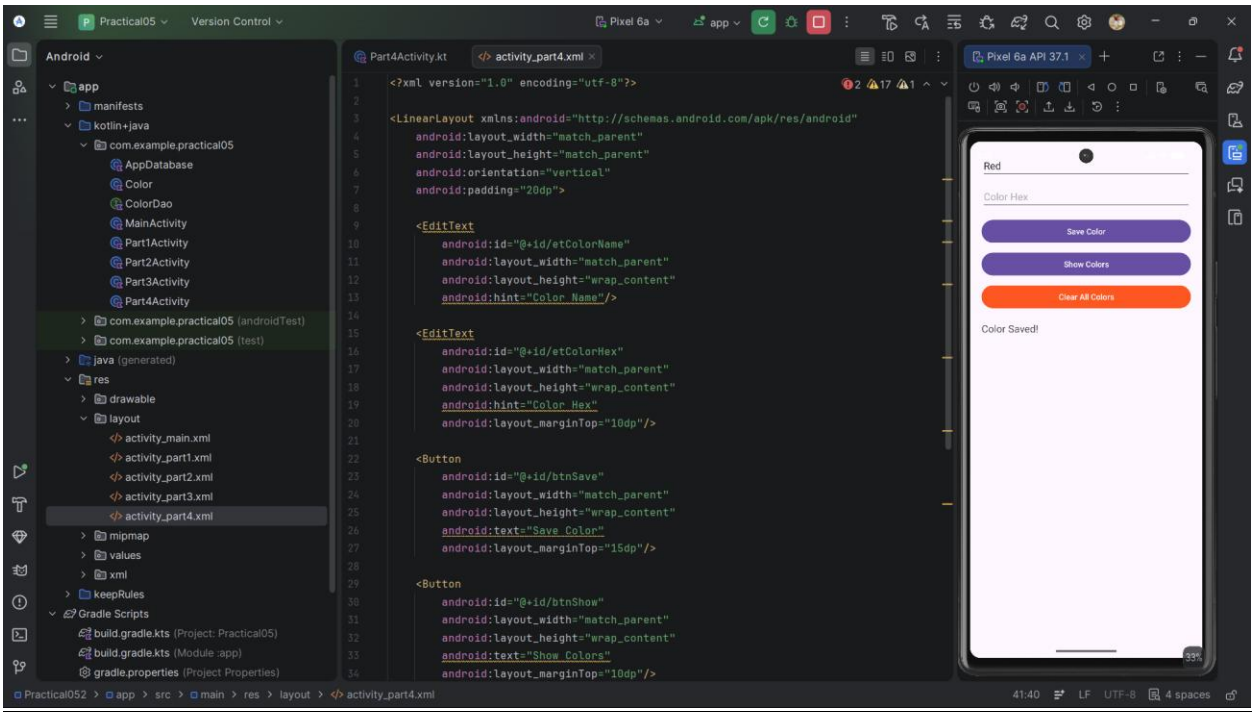

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build.gradle.kts (Project: Practical05)
build.gradle.kts (Module :app)
gradle.properties (Project Properties)
class Part4Activity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
val prefs = getSharedPreferences(name = "ColorPrefs", mode = Context.MODE_PRIVATE)
// Auto-fill last color name
etName.setText(prefs.getString(p0 = "lastColor", p1 = ""))
btnSave.setOnClickListener {
val name = etName.text.toString()
val hex = etHex.text.toString()
val color = Color(
name = name,
hex = hex
)
database.colorDao().insert(color)
// Save last entered color name
prefs.edit()
.putString(p0 = "lastColor", p1 = name)
.apply()
tvColors.text = "Color Saved!"
}
btnShow.setOnClickListener {
val list = database.colorDao().getAll()
var result = ""
for (color in list) {
result += "${color.name} - ${color.hex}\n"
}
tvColors.text = result
}
OFF
Practical052 › app › src › main › java › com › example › practical05 › Part4Activity › database
13:47
LF
UTF-8
4 spaces

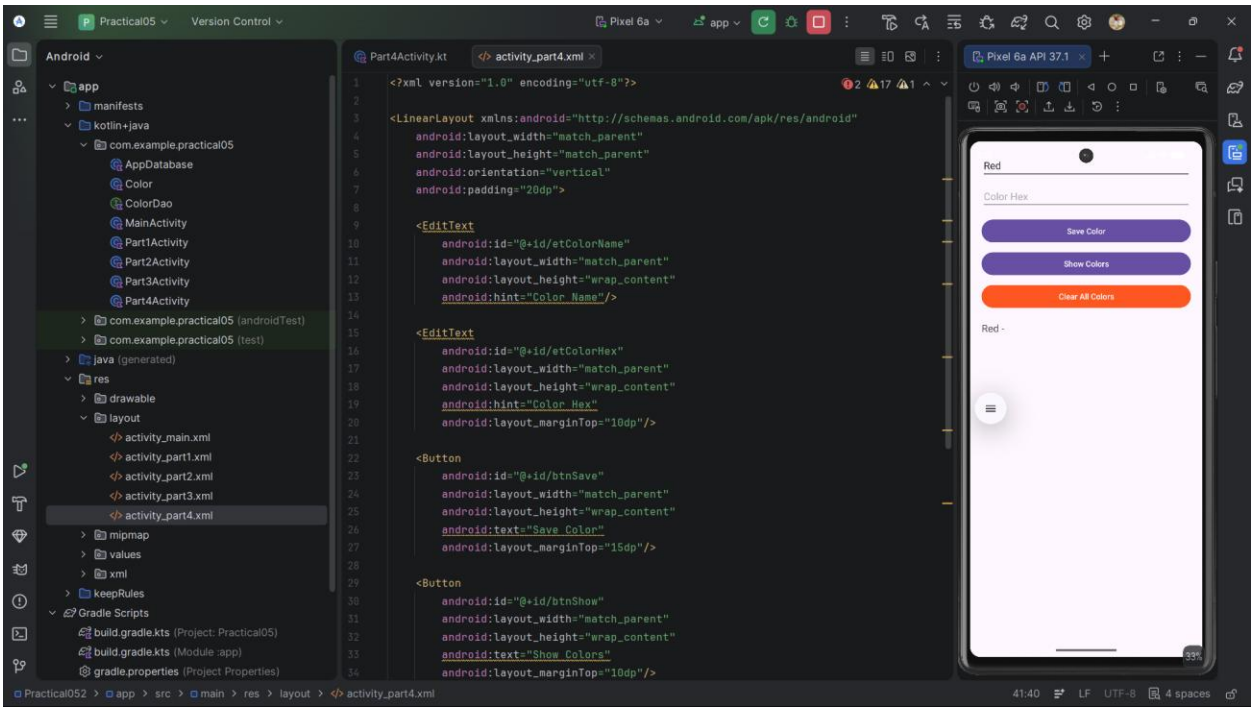

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gradle.properties (Project Properties)
class Part4Activity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
btnSave.setOnClickListener {
// Save last entered color name
prefs.edit()
.putString(p0 = "lastColor", p1 = name)
.apply()
tvColors.text = "Color Saved!"
}
btnShow.setOnClickListener {
val list = database.colorDao().getAll()
var result = ""
for (color in list) {
result += "${color.name} - ${color.hex}\n"
}
tvColors.text = result
}
btnClearAll.setOnClickListener {
database.colorDao().deleteAll()
// Also clear the last saved name from prefs
prefs.edit().remove(p0 = "lastColor").apply()
etName.text.clear()
tvColors.text = "All Cleared!"
}
}
}
OFF
Practical052 › app › src › main › java › com › example › practical05 › Part4Activity › database
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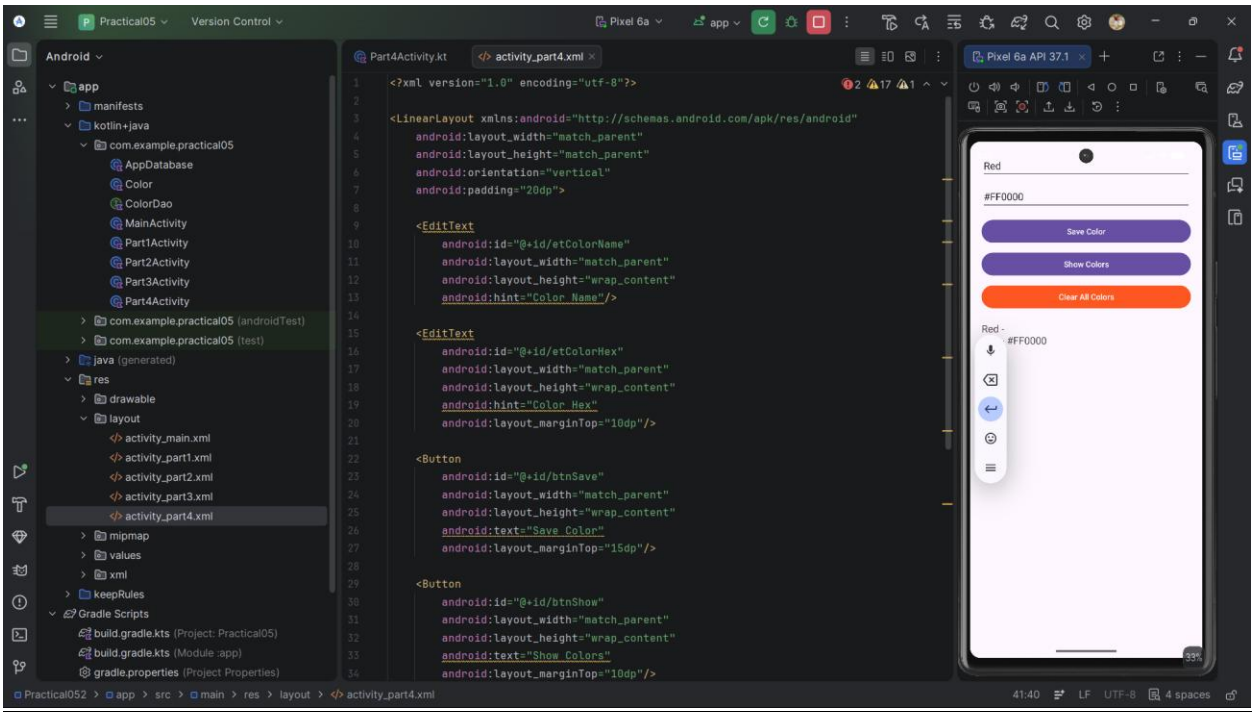

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build.gradle.kts (Module :app)
gradle.properties (Project Properties)
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp">
<EditText
android:id="@+id/etColorName"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Color Name"/>
<EditText
android:id="@+id/etColorHex"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Color Hex"
android:layout_marginTop="10dp"/>
<Button
android:id="@+id/btnSave"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Save Color"
android:layout_marginTop="15dp"/>
<Button
android:id="@+id/btnShow"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Show Colors"
android:layout_marginTop="10dp"/>
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<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
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android:layout_height="wrap_content"
android:text="Save Color"
android:layout_marginTop="15dp"/>
<Button
android:id="@+id/btnShow"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Show Colors"
android:layout_marginTop="10dp"/>
<Button
android:id="@+id/btnClearAll"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Clear All Colors"
android:layout_marginTop="10dp"
android:backgroundTint="#FF5722"/>
<TextView
android:id="@+id/tvColors"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="18sp"
android:layout_marginTop="20dp"/>
</LinearLayout>
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android:text="Show Colors"
android:layout_marginTop="10dp"/>
Color Name
Color Hex
Save Color
Show Colors
Clear All Colors
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android:layout_marginTop="10dp"/>
Red
Color Hex
Save Color
Show Colors
Clear All Colors
Color Saved!
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Red
Color Hex
Save Color
Show Colors
Clear All Colors
Red -
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Red
#FF0000
Save Color
Show Colors
Clear All Colors
Red -
#FF0000
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